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(54) **NETWORK DATA ANALYTICS FUNCTION ENHANCEMENTS FOR ARTIFICIAL INTELLIGENCE/MACHINE LEARNING-BASED LOCATION SERVICES POSITIONING**

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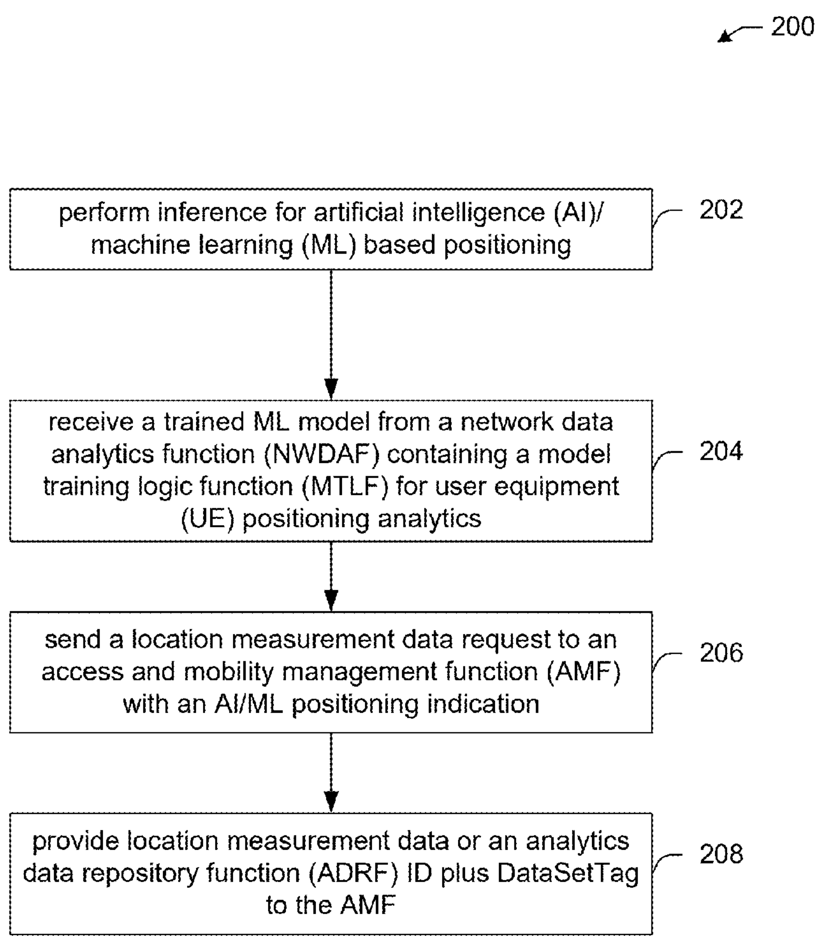
(57) **ABSTRACT**

This disclosure describes systems, methods, and devices related to enhanced NWDAF. A device may perform inference for artificial intelligence (AI)/machine learning (ML) based positioning. The device may receive a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics. The device may send a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication. The device may provide location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF.



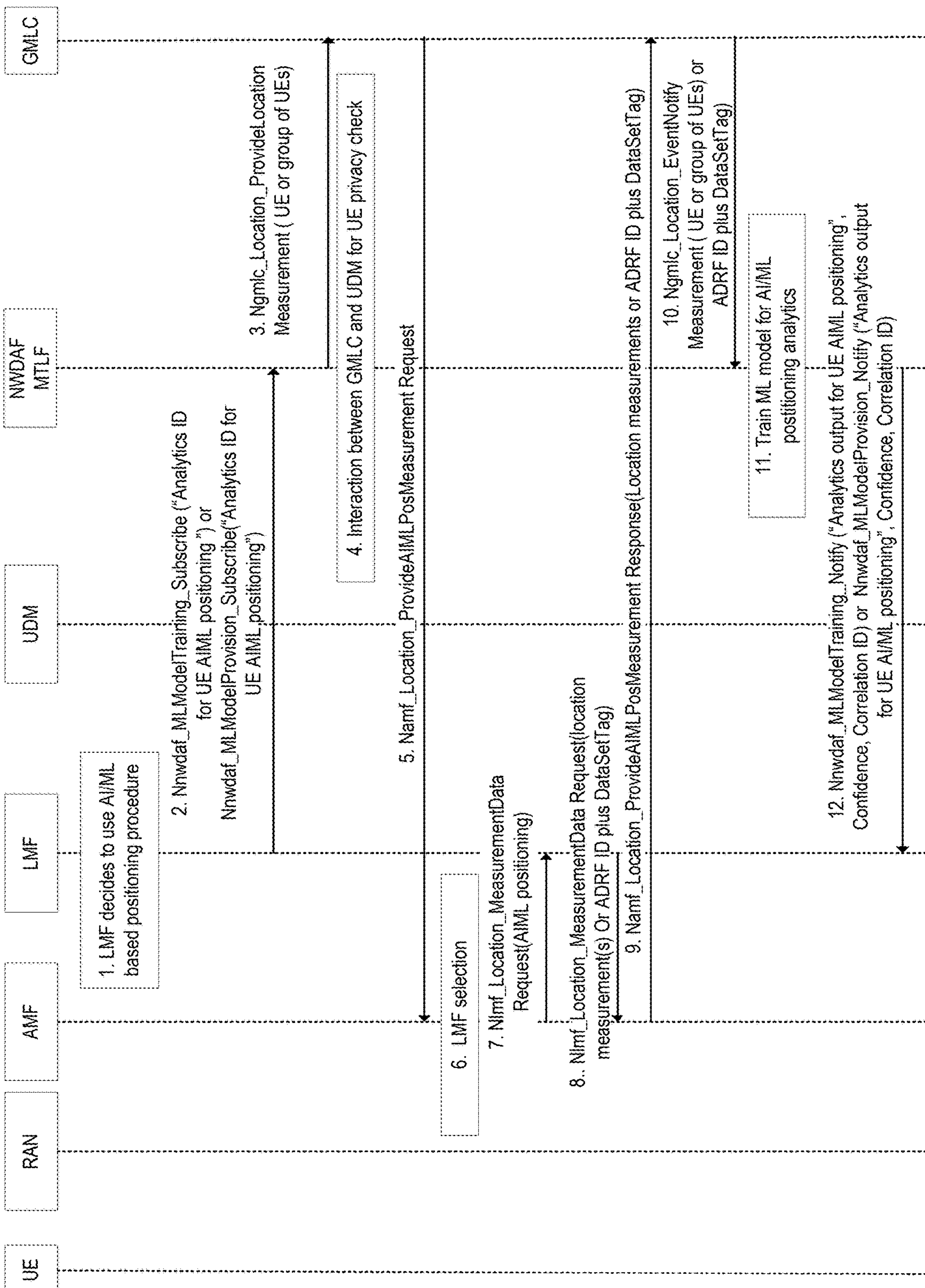

UE
RAN
AMF
LMF
UDM
NWDAF
MTLF
GMLC
1. LMF decides to use AI/ML based positioning procedure
2. Nnwdaf_MLModelTraining_Subscribe ("Analytics ID for UE AIML positioning") or Nnwdaf_MLModelProvision_Subscribe("Analytics ID for UE AIML positioning")
3. Ngmlc_Location_ProvideLocation Measurement (UE or group of UEs)
4. Interaction between GMLC and UDM for UE privacy check
5. Namf_Location_ProvideAIMLPosMeasurement Request
6. LMF selection
7. Nlmf_Location_MeasurementData Request(AIML positioning)
8.. Nlmf_Location_MeasurementData Request(location measurement(s) Or ADRF ID plus DataSetTag)
9. Namf_Location_ProvideAIMLPosMeasurement Response(Location measurements or ADRF ID plus DataSetTag)
10. Ngmlc_Location_EventNotify Measurement (UE or group of UEs) or ADRF ID plus DataSetTag)
11. Train ML model for AI/ML positioning analytics
12. Nnwdaf_MLModelTraining_Notify ("Analytics output for UE AIML positioning", Confidence, Correlation ID) or Nnwdaf_MLModelProvision_Notify ("Analytics output for UE AI/ML positioning", Confidence, Correlation ID)

FIG. 1

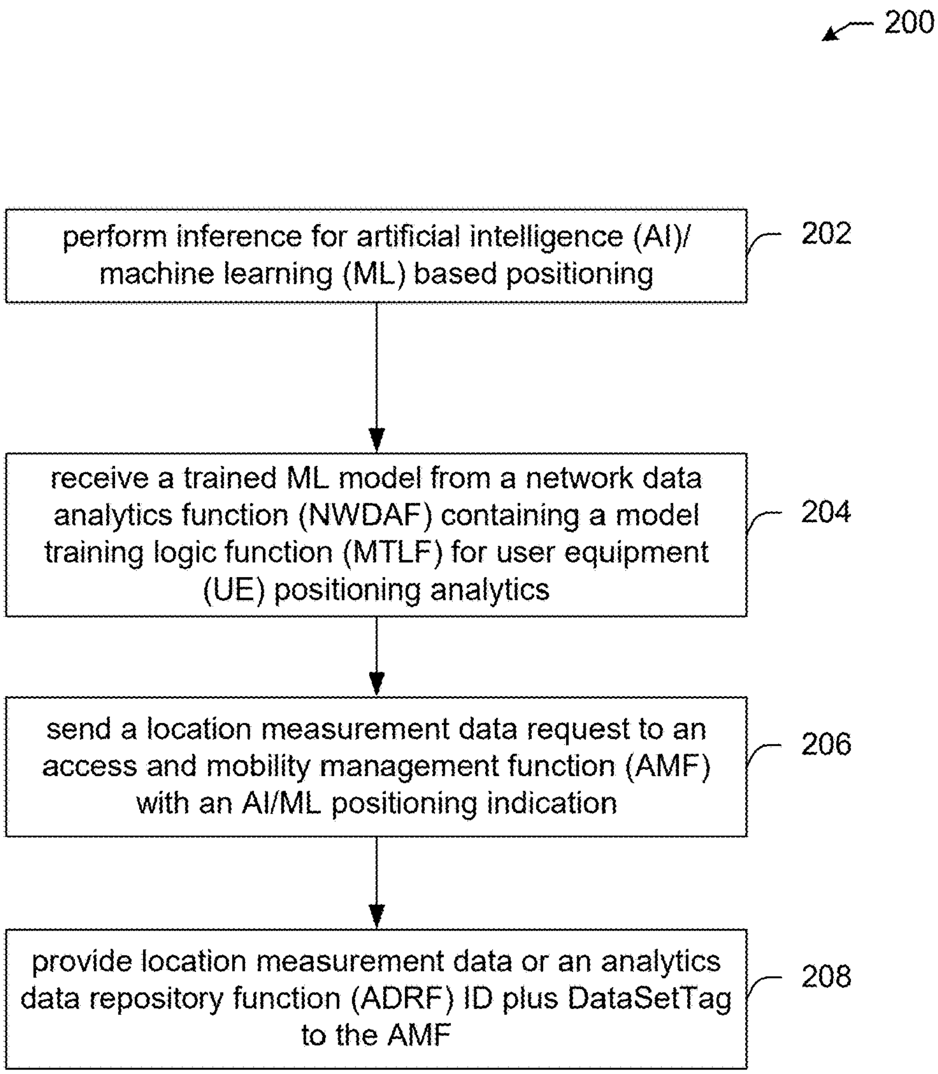

200
perform inference for artificial intelligence (AI)/ machine learning (ML) based positioning
202
receive a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics
204
send a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication
206
provide location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF
208

FIG. 2

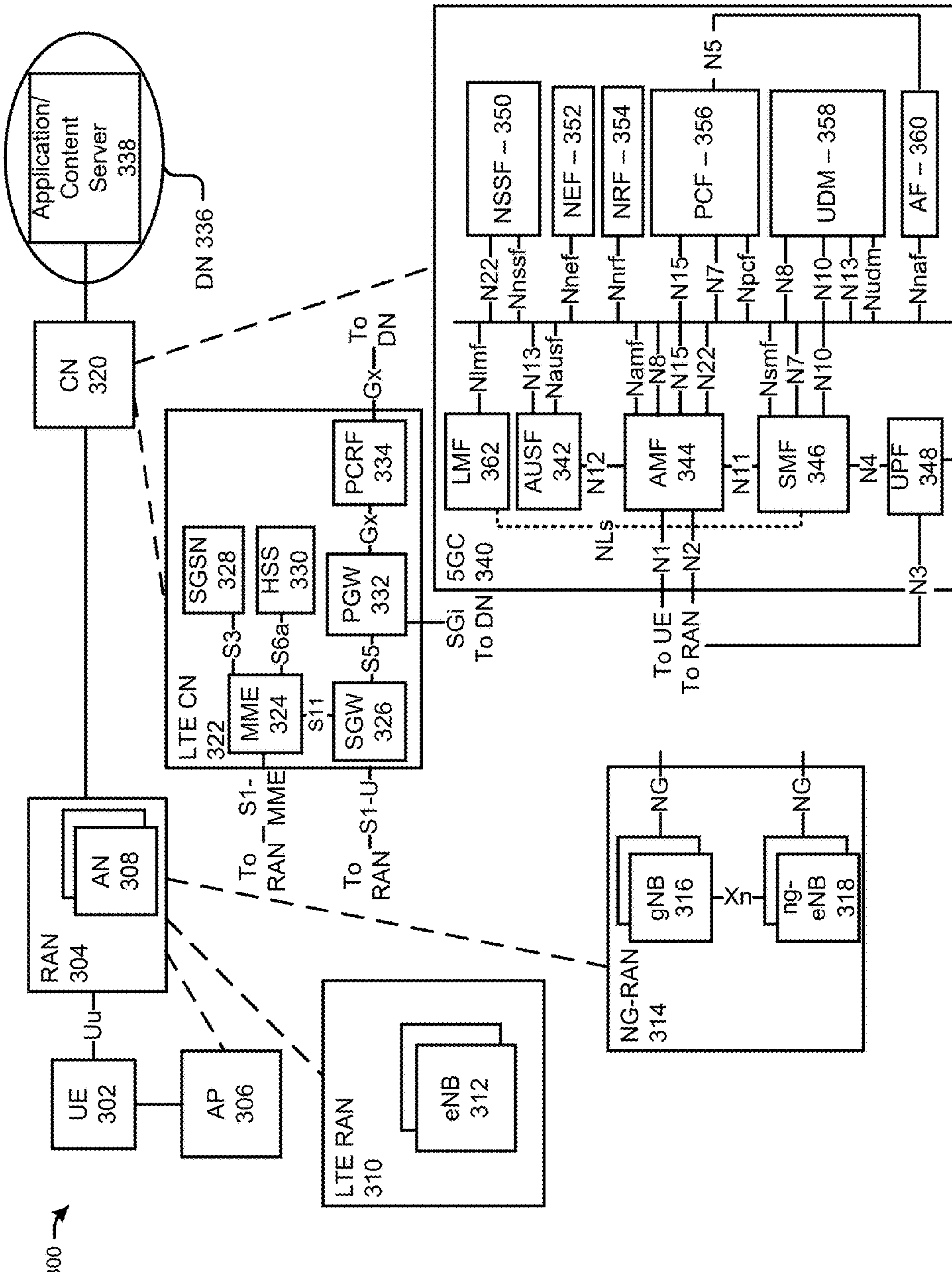
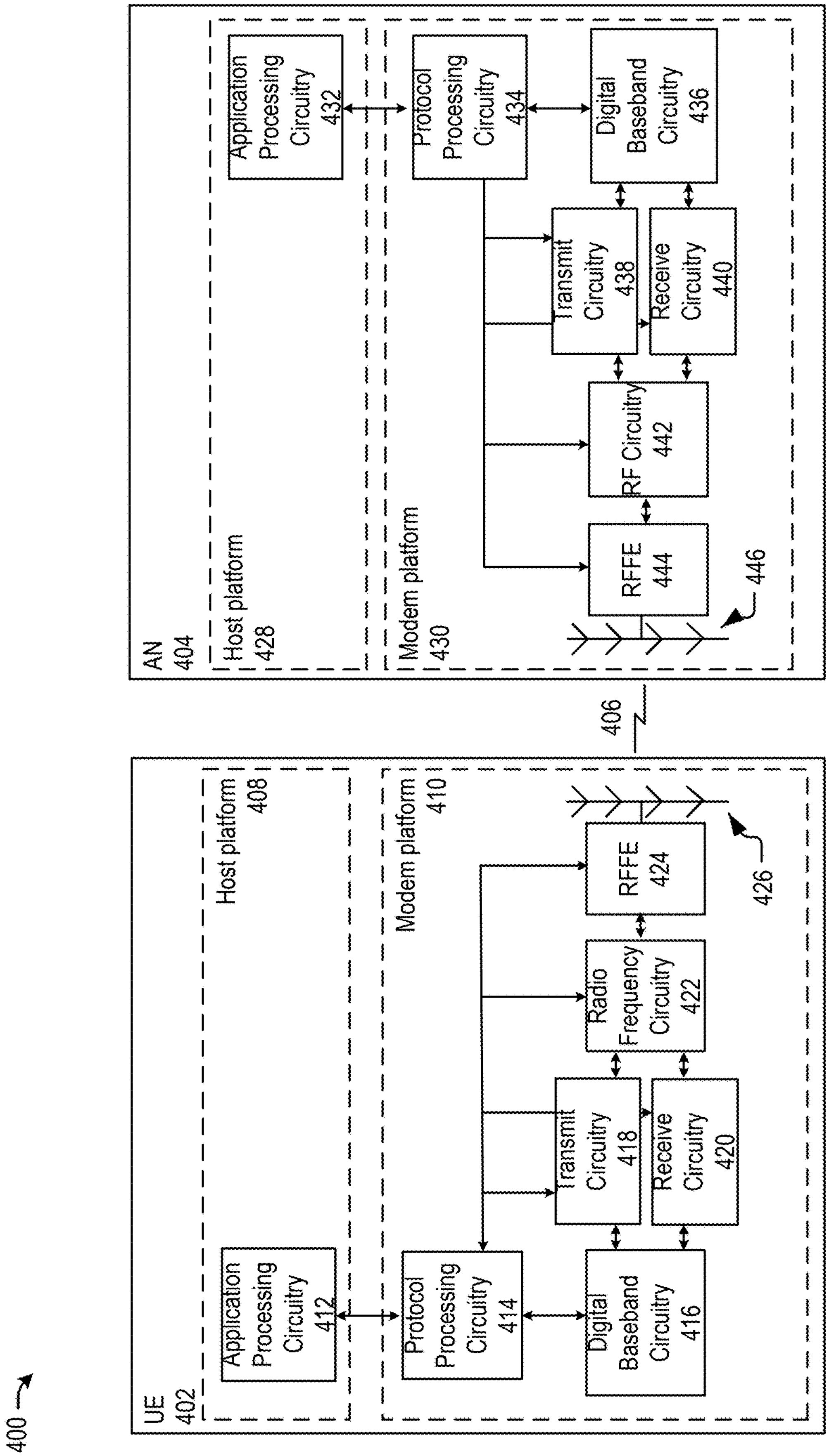

300
UE 302
Uu
AP 306
RAN 304
AN 308
LTE RAN 310
eNB 312
NG-RAN 314
gNB 316
Xn
ng-eNB 318
NG
NG
CN 320
LTE CN 322
MME 324
SGW 326
SGSN 328
HSS 330
PGW 332
PCRF 334
S3
S6a
S11
S5
Gx
Gx
To DN
SGi To DN
To RAN
S1-MME
To RAN
S1-U
DN 336
Application/Content Server 338
5GC 340
AUSF 342
AMF 344
SMF 346
UPF 348
NSSF -- 350
NEF -- 352
NRF -- 354
PCF -- 356
UDM -- 358
AF -- 360
LMF 362
Nlmf
N13
Nausf
Namf
N8
N15
N22
Nsmf
N7
N10
N22
Nnssf
Nnef
Nnrf
N15
N7
Npcf
N8
N10
N13
Nudm
Nnaf
N5
N12
N11
N4
NLs
N1
N2
N3
To UE
To RAN
N6 To DN

FIG. 3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UE
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Host platform
408
Application
Processing
Circuitry
412
Modem platform
410
Protocol
Processing
Circuitry
414
Digital
Baseband
Circuitry
416
Transmit
Circuitry
418
Receive
Circuitry
420
Radio
Frequency
Circuitry
422
RFFE
424
426
406
AN
404
Host platform
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Application
Processing
Circuitry
432
Modem platform
430
Protocol
Processing
Circuitry
434
Digital
Baseband
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Transmit
Circuitry
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Receive
Circuitry
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RF Circuitry
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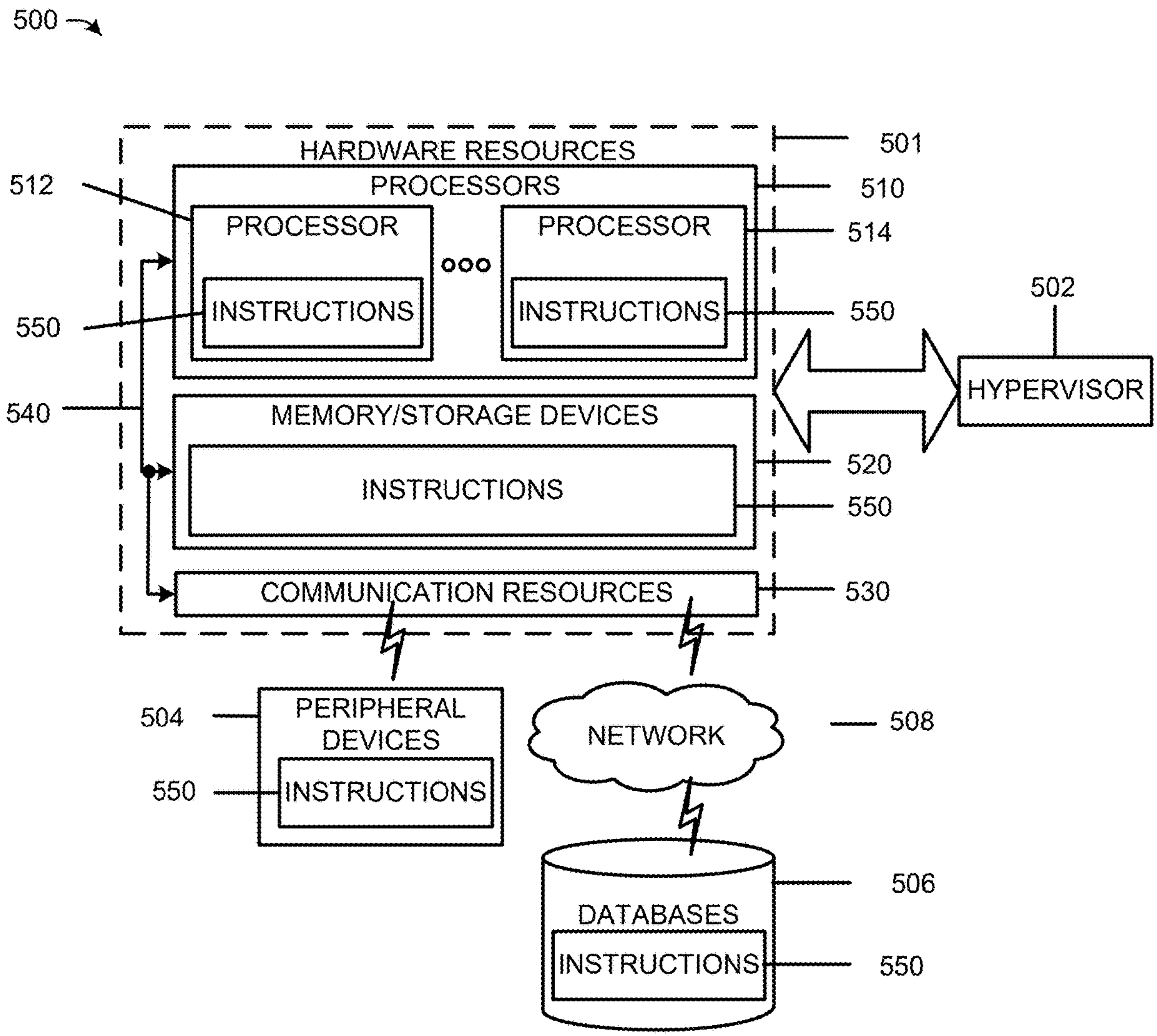
RFFE
444
446

FIG. 4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HARDWARE RESOURCES
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PROCESSORS
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PROCESSOR
PROCESSOR
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INSTRUCTIONS
INSTRUCTIONS
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550
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HYPERVISOR
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MEMORY/STORAGE DEVICES
INSTRUCTIONS
520
550
COMMUNICATION RESOURCES
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504
PERIPHERAL DEVICES
550
INSTRUCTIONS
NETWORK
508
506
DATABASES
INSTRUCTIONS
550

FIG. 5

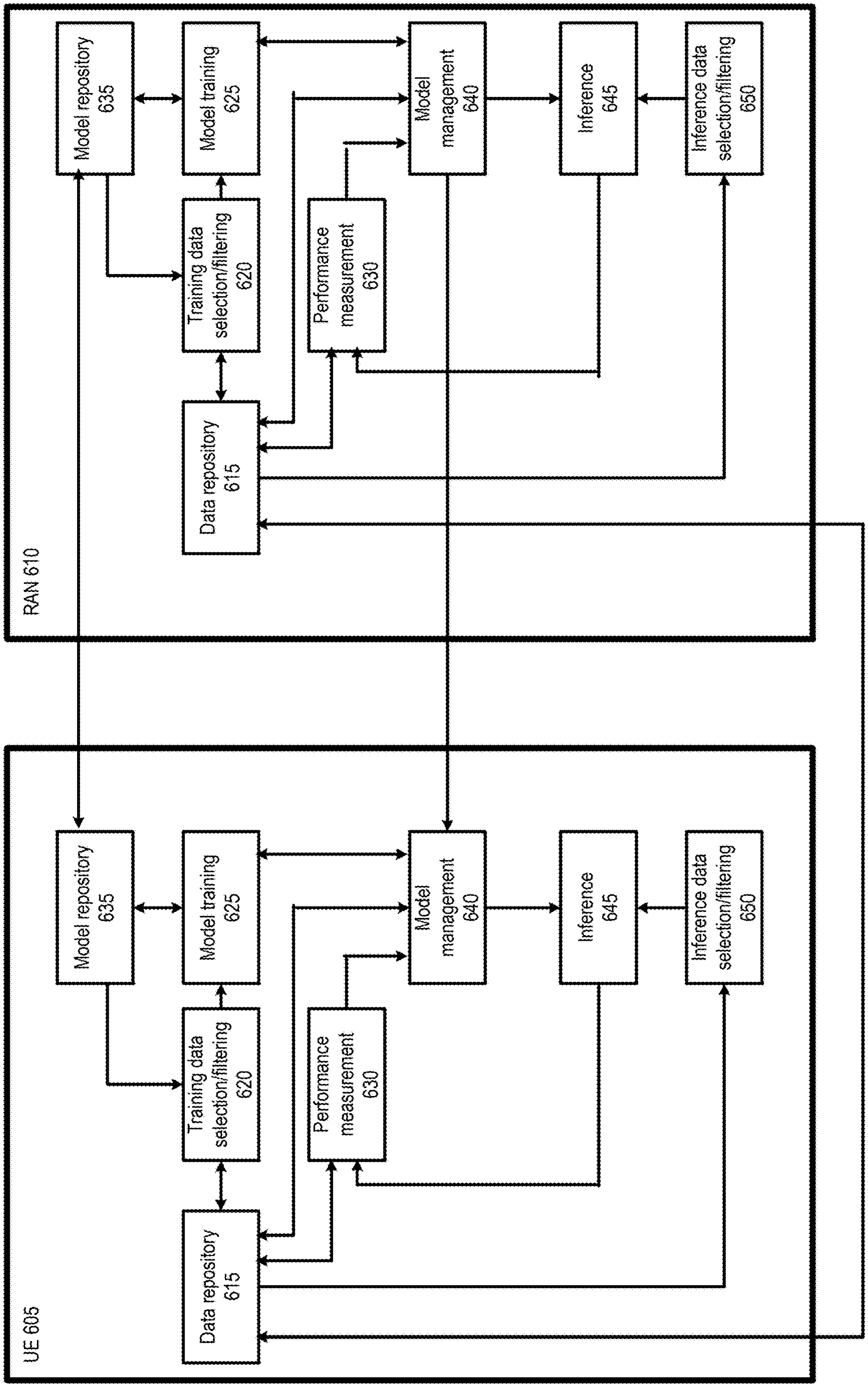
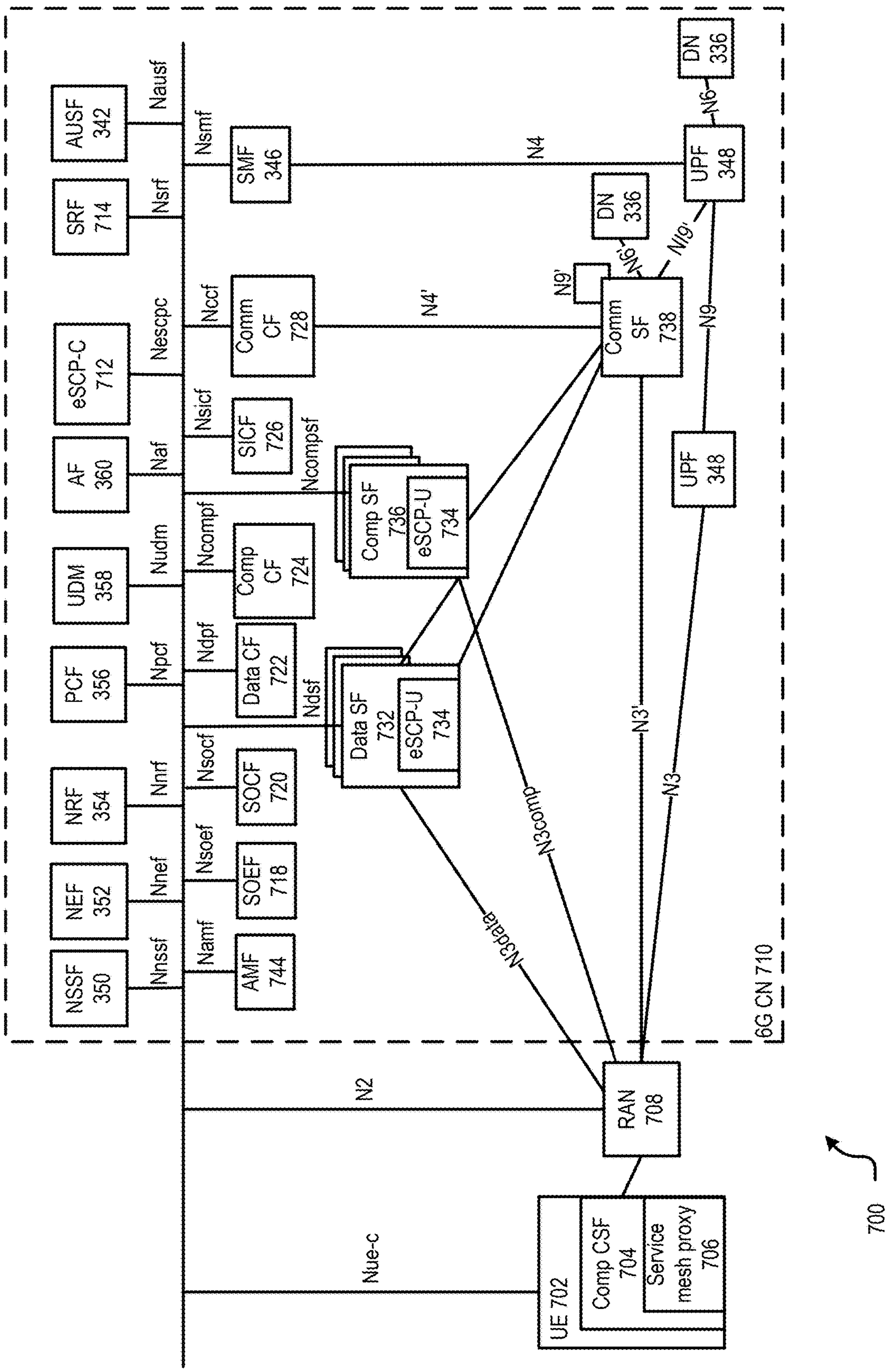

RAN 610
Model repository 635
Model training 625
Training data selection/filtering 620
Performance measurement 630
Model management 640
Inference 645
Inference data selection/filtering 650
Data repository 615
UE 605
Model repository 635
Model training 625
Training data selection/filtering 620
Performance measurement 630
Model management 640
Inference 645
Inference data selection/filtering 650
Data repository 615

FIG. 6



NSSF 350
NEF 352
NRF 354
PCF 356
UDM 358
AF 360
eSCP-C 712
SRF 714
AUSF 342
Nnssf
Nnef
Nnrf
Npcf
Nudm
Naf
Nescpc
Nsrf
Nausf
Namf
Nsoef
Nsocf
Ndsf
Ndpf
Ncompf
Ncompsf
Nsicf
Nccf
Nsmf
AMF 744
SOEF 718
SOCF 720
Data CF 722
Comp CF 724
SICF 726
Comm CF 728
SMF 346
Data SF 732
eSCP-U 734
Comp SF 736
eSCP-U 734
Comm SF 738
N9'
DN 336
DN 336
UPF 348
UPF 348
N4
N4'
N6'
N9'
N6
N9
N3'
N3
N3comp
N3data
6G CN 710
N2
RAN 708
Nue-c
UE 702
Comp CSF 704
Service mesh proxy 706
700

FIG. 7

NETWORK DATA ANALYTICS FUNCTION ENHANCEMENTS FOR ARTIFICIAL INTELLIGENCE/MACHINE LEARNING-BASED LOCATION SERVICES POSITIONING

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefit of U.S. Provisional Application No. 63/572,124, filed Mar. 29, 2024, the disclosure of which is incorporated herein by reference as if set forth in full.

BACKGROUND

[0002] Wireless networks are essential for modern communication, supporting diverse devices and applications. As data demands grow, these networks must enhance performance and reliability. Key advancements focus on optimizing data handling and network management.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** depicts an illustrative schematic diagram for enhanced network data analytics function (NWDAF), in accordance with one or more example embodiments of the present disclosure.

[0004] FIG. **2** illustrates a flow diagram of illustrative process for an illustrative enhanced NWDAF system, in accordance with one or more example embodiments of the present disclosure.

[0005] FIG. **3** illustrates an example network architecture, in accordance with one or more example embodiments of the present disclosure.

[0006] FIG. **4** schematically illustrates a wireless network, in accordance with one or more example embodiments of the present disclosure.

[0007] FIG. **5** illustrates components of a computing device, in accordance with one or more example embodiments of the present disclosure.

[0008] FIG. **6** illustrates a network in accordance with various embodiments.

[0009] FIG. **7** illustrates a simplified block diagram of artificial (AI)-assisted communication between a user equipment (UE) and a radio access network (RAN), in accordance with various embodiments.

DETAILED DESCRIPTION

[0010] The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, algorithm, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

[0011] In the third generation partnership project (3GPP) release-19 (Rel-19) specifications, enhancements to location services (LCS) to support Direct artificial intelligence/machine learning (AI/ML)-based positioning was agreed to be studied. Specifically, aspects to be considered may have included one or more of the following:

[0012] Study whether and how an AI/ML model for Direct AI/ML positioning (i.e. case 2b/3b) is handled:

[0013] Which entity trains the model for Direct AI/ML positioning and if the entity that train the model and the consumer are different, how the Model consumer gets the trained AI/ML model.

[0014] How the Model consumer uses the trained model to perform inference and/or derive UE position.

[0015] Define procedures for data collection with objective to train AI/ML models for Direct AI/ML positioning.

[0016] Whether and how to support Direct AI/ML positioning with additional fifth generation core (5GC) enhancements.

[0017] How to monitor model performance for ML models used for Direct AI/ML based positioning.

[0018] Example embodiments of the present disclosure relate to systems, methods, and devices for network data analytics function (NWDAF) machine learning (ML) model training and data collection procedure for location services (LCS) architecture enhancements to support direct artificial intelligence (AI)/ML based positioning.

[0019] Embodiments herein relate to fifth generation system (5GS) architectural enhancement options to support AI/ML-based user equipment (UE) positioning with NWDAF model training logical function (MTLF) training the ML model required by the LCS for model inference and supported data collection procedure. Embodiments may be similar to those described in, for example, the ML model training procedure to support UE positioning analytics.

[0020] In one or more embodiments, a system may comprise one or more components, which may include one or more of: apparatus, device, user equipment (UE), gNodeB, and/or other network elements. At its most basic configuration, the system includes one or more processors, and/or memory, and/or instructions. The processor(s) may be implemented using general-purpose microprocessors, and/or digital signal processors (DSPs), and/or field-programmable gate arrays (FPGAs), and/or other suitable computational entities capable of performing calculations or manipulations of information. The memory may include RAM, and/or ROM, and/or flash memory, and/or other storage media suitable for storing instructions and/or data necessary for system operation. These components, individually or in combination, enable the execution of processes that facilitate communication and/or functionality within the system.

[0021] The above descriptions are for purposes of illustration and are not meant to be limiting. Numerous other examples, configurations, processes, algorithms, etc., may exist, some of which are described in greater detail below. Example embodiments will now be described with reference to the accompanying figures.

[0022] FIG. **1** depicts an illustrative schematic diagram for enhanced NWDAF, in accordance with one or more example embodiments of the present disclosure.

[0023] When the location management function (LMF) uses AI/ML-based positioning, the following procedure may be followed to request the trained ML model for UE positioning analytics by the LMF and the corresponding data collection method required at the NWDAF containing MTLF for ML model training.

[0024] 1. When the LMF determines to use AI/ML based positioning method, the LMF may get a trained ML model from NWDAF containing MTLF for UE AI/ML positioning analytics. This approach allows the LMF to leverage advanced analytics for more accurate positioning, such as using real-time data to improve location precision. The LMF decision to get a trained ML model from NWDAF may be based on LMF implementation or based on ML model accuracy monitoring requirement by the service consumer of the trained ML model for UE AIML positioning analytics. This ensures that the chosen model meets specific performance criteria, enhancing user experience by providing reliable location data.

[0025] 2. The LMF sends Nnwdaf_MLModelProvision_Subscribe or Nnwdaf_MLModelTraining_Subscribe for UE AIML positioning analytics as using the Nnwdaf services.

[0026] 3. Based on the ML model training request from the service consumer (in this case the LMF), the NWDAF MTLF initiates the data collection required to training train the ML model for UE AIML positioning analytics from the GMLC using Ngmlc_Location_ProvideLocationMeasurement request for a UE or group of UEs which may be determined based on the ML Model Filter Information in step 2.

[0027] 4. The GMLC invokes a Nudm_SDM_Get (LCS privacy, SUPI) service operation towards the UDM to get the UE LCS privacy profile of the target UE. The GMLC invokes a Nudm_UECM_Get service operation towards the UDM of the target UE with SUPI of this UE. The UDM returns the current serving AMF ID to the GMLC.

[0028] 5. Based on the serving AMF ID in step 4, the GMLC sends Namf_Location_ProvideAIMLPosMeasurement Request to the AMF.

[0029] 6. The AMF performs the LMF selection to select the LMF(s) for the UE (or group of UEs).

[0030] 7. The AMF sends the Nlmf_Location_MeasurementDataRequest to the LMF(s) selected in step 6 with the AIMLpositioning indication to indicate that the measurement data request is for AIML positioning method. This indication is required for the case when the positioning measurement parameters that is to be collected for AIML positioning method is different from the current parameters defined for other positioning methods in RAN.

[0031] 8. The LMF sends the Nlmf_Location_MeasurementData response to the AMF with the requested location measurement data or the ADRF ID plus DataSetTag where the AIML based measurement data is stored. The LMF may send the ADRF ID plus DataSetTag if the LMF stores the measurement data collected during ML model inference or stored in ADRF as historical data collection for UE or group of UEs. The LMF stores the measurement data collected during ML model inference in the ADRF, then the ADRF ID and DataSetTag can be sent to the consumer (NWDAF MTLF) to retrieve the input data required for ML model training. If the requested measurement data requested is not available at the LMF or ADRF, the LMF will collect different measurement data from UE or RAN for model training to obtain UE location measurement data.

[0032] If the LMF stores the measurement data collected during ML model inference in the ADRF, then the ADRF ID and DataSetTag can be sent to the consumer (NWDAF MTLF) to retrieve the input data required for ML model training. If the LMF does not store the measurement data collected during ML model inference, then when there is a request for input data required for ML model training then the LMF will initiate a location+positioning request for a UE.

[0033] If the LMF stores the measurement data collected during ML model inference in the ADRF, then the ADRF ID and DataSetTag can be sent to the consumer (NWDAF MTLF) to retrieve the input data required for ML model training. This process ensures efficient data management and accessibility, allowing for seamless integration and updates to the ML model. If the LMF does not store the measurement data collected during ML model inference, then when there is a request for input data required for ML model training, the LMF will initiate a location+positioning request for a UE. This approach ensures that necessary data is collected on-demand, maintaining the relevance and accuracy of the ML model.

[0034] 9. The AMF sends the Namf_Location_ProvideAIMLPosMeasurement Response to the GMLC with the location measurement data or the ADRF ID plus DataSetTag where the AIML based measurement data is stored.

[0035] 10. GMLC may aggregate one or more UE location measurement data received or ADRF ID plus DataSetTag reports in each message sent to NWDAF containing MTLF.

[0036] 11. The NWDAF containing MTLF trains the ML model for AIML positioning analytics based on the data collected in step 10 or data stored in the ADRF ID plus DataSetTag by the LMF.

[0037] 12. The NWDAF containing MTLF sends the Nnwdaf_MLModelProvision_Notify or Nnwdaf_MLModelTraining_Notify for AIML positioning analytics to the LMF as requested in step 2.

[0038] In one or more embodiments, the device may support AI/ML-based positioning in 5GS by using LMF to perform inference for AI/ML positioning, while NWDAF containing MTLF may handle ML model training for UE AI/ML positioning analytics in the 5GC. The LMF may decide to obtain a trained ML model from NWDAF based on implementation or accuracy monitoring requirements by the service consumer. In this role, the LMF may send the Nnwdaf_MLModelTraining_Subscribe for UE AI/ML positioning analytics. The NWDAF MTLF may initiate data collection for ML model training from the GMLC using the Ngmlc_Location_ProvideLocationMeasurement request for a UE or group of UEs, determined by ML Model Filter Information. The GMLC may send a Namf_Location_ProvideAIMLPosMeasurement Request to the AMF(s) serving the UE or group of UEs. The AMF may then send the Nlmf_Location_MeasurementDataRequest to the LMF(s) with an AIML positioning indication. The LMF may respond with the Nlmf_Location_MeasurementData to the AMF, providing either the requested location measurement data or the ADRF ID plus DataSetTag where the AIML data is stored. If the LMF stores measurement data during ML model inference, it may send the ADRF ID plus DataSetTag to the NWDAF MTLF for data retrieval. If the requested data is unavailable, the LMF may collect different measurement data from UE or RAN for ML model training. The AMF may send the Namf_Location_ProvideAIMLPosMeasurement Response to the GMLC with the location data or ADRF ID plus DataSetTag. The GMLC may aggregate UE location data or ADRF ID plus DataSetTag reports in messages sent to NWDAF containing MTLF. Finally, the NWDAF containing MTLF may send the Nnwdaf_MLModelTraining_Notify for AI/ML positioning analytics to the LMF.

[0039] It is understood that the above descriptions are for the purposes of illustration and are not meant to be limiting.

[0040] In some embodiments, the electronic device(s), network(s), system(s), chip(s) or component(s), or portions or implementations thereof, of the figures herein may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof. One such process is depicted in FIG. **2**.

[0041] For example, the process may include, at **202**, performing inference for artificial intelligence (AI)/machine learning (ML) based positioning.

[0042] The process further includes, at **204**, receiving a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics.

[0043] The process further includes, at **206**, sending a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication.

[0044] The process further includes, at **208**, providing location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF.

[0045] For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

[0046] It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

[0047] The figures described below illustrate illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

[0048] FIG. **3** illustrates an example network architecture **300** according to various embodiments. The network **300** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

[0049] The network **300** includes a UE **302**, which is any mobile or non-mobile computing device designed to communicate with a RAN **304** via an over-the-air connection. The UE **302** is communicatively coupled with the RAN **304** by a Uu interface, which may be applicable to both LTE and NR systems. Examples of the UE **302** include, but are not limited to, a smartphone, tablet computer, wearable computer, desktop computer, laptop computer, in-vehicle infotainment system, in-car entertainment system, instrument cluster, head-up display (HUD) device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, machine-to-machine (M2M), device-to-device (D2D), machine-type communication (MTC) device, Internet of Things (IoT) device, and/or the like. The network **300** may include a plurality of UEs **302** coupled directly with one another via a D2D, ProSe, PC5, and/or sidelink (SL) interface. These UEs **302** may be M2M/D2D/MTC/IoT devices and/or vehicular systems that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc. The UE **302** may perform blind decoding attempts of SL channels/links according to the various embodiments herein.

[0050] In some embodiments, the UE **302** may additionally communicate with an AP **306** via an over-the-air (OTA) connection. The AP **306** manages a WLAN connection, which may serve to offload some/all network traffic from the RAN **304**. The connection between the UE **302** and the AP **306** may be consistent with any IEEE 802.11 protocol. Additionally, the UE **302**, RAN **304**, and AP **306** may utilize cellular-WLAN aggregation/integration (e.g., LWA/LWIP). Cellular-WLAN aggregation may involve the UE **302** being configured by the RAN **304** to utilize both cellular radio resources and WLAN resources.

[0051] The RAN **304** includes one or more access network nodes (ANs) **308**. The ANs **308** terminate air-interface(s) for the UE **302** by providing access stratum protocols including RRC, PDCP, RLC, MAC, and PHY/L1 protocols. In this manner, the AN **308** enables data/voice connectivity between CN **320** and the UE **302**. The ANs **308** may be a macrocell base station or a low power base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells; or some combination thereof. In these implementations, an AN **308** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, etc.

[0052] One example implementation is a "CU/DU split" architecture where the ANs **308** are embodied as a gNB-Central Unit (CU) that is communicatively coupled with one or more gNB-Distributed Units (DUs), where each DU may be communicatively coupled with one or more Radio Units (RUs) (also referred to as RRHs, RRUs, or the like) (see e.g., 3GPP TS 38.401 v16.1.0 (2020 March)). In some implementations, the one or more RUs may be individual RSUs. In some implementations, the CU/DU split may include an ng-eNB-CU and one or more ng-eNB-DUs instead of, or in addition to, the gNB-CU and gNB-DUs, respectively. The ANs **308** employed as the CU may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network including a virtual Base Band Unit (BBU) or BBU pool, cloud RAN (CRAN), Radio Equipment Controller (REC), Radio Cloud Center (RCC), centralized RAN (C-RAN), virtualized RAN (vRAN), and/or the like (although these terms may refer to different implementation concepts). Any other type of architectures, arrangements, and/or configurations can be used.

[0053] The plurality of ANs may be coupled with one another via an X2 interface (if the RAN **304** is an LTE RAN or Evolved Universal Terrestrial Radio Access Network (E-UTRAN) **310**) or an Xn interface (if the RAN **304** is a NG-RAN **314**). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

[0054] The ANs of the RAN **304** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **302** with an air interface for network access. The UE **302** may be simultaneously connected with a plurality of cells provided by the same or different ANs **308** of the RAN **304**. For example, the UE **302** and RAN **304** may use carrier aggregation to allow the UE **302** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN **308** may be a master node that provides an MCG and a second AN **308** may be secondary node that provides an SCG. The first/second ANs **308** may be any combination of eNB, gNB, ng-eNB, etc.

[0055] The RAN **304** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Prior to accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

[0056] In V2X scenarios the UE **302** or AN **308** may be or act as a roadside unit (RSU), which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by: a UE may be referred to as a "UE-type RSU"; an eNB may be referred to as an "eNB-type RSU"; a gNB may be referred to as a "gNB-type RSU"; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

[0057] In some embodiments, the RAN **304** may be an E-UTRAN **310** with one or more eNBs **312**. The an E-UTRAN **310** provides an LTE air interface (Uu) with the following characteristics: SCS of 15 kHz; CP-OFDM waveform for DL and SC-FDMA waveform for UL; turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operating on sub-6 GHz bands.

[0058] In some embodiments, the RAN **304** may be an next generation (NG)-RAN **314** with one or more gNB **316** and/or on or more ng-eNB **318**. The gNB **316** connects with 5G-enabled UEs **302** using a 5G NR interface. The gNB **316** connects with a 5GC **340** through an NG interface, which includes an N2 interface or an N3 interface. The ng-eNB **318** also connects with the 5GC **340** through an NG interface, but may connect with a UE **302** via the Uu interface. The gNB **316** and the ng-eNB **318** may connect with each other over an Xn interface.

[0059] In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **314** and a UPF **348** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **314** and an AMF **344** (e.g., N2 interface).

[0060] The NG-RAN **314** may provide a 5G-NR air interface (which may also be referred to as a Uu interface) with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS, but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH; and tracking reference signal for time tracking. The 5G-NR air interface may operating on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include an SSB that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

[0061] The 5G-NR air interface may utilize BWPs for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **302** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **302**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **302** with different amount of frequency resources (e.g., PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with small traffic load while allowing power saving at the UE **302** and in some cases at the gNB **316**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic load.

[0062] The RAN **304** is communicatively coupled to CN **320** that includes network elements and/or network functions (NFs) to provide various functions to support data and telecommunications services to customers/subscribers (e.g., UE **302**). The components of the CN **320** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **320** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **320** may be referred to as a network slice, and a logical instantiation of a portion of the CN **320** may be referred to as a network sub-slice.

[0063] The CN **320** may be an LTE CN **322** (also referred to as an Evolved Packet Core (EPC) **322**). The EPC **322** may include MME **324**, SGW **326**, SGSN **328**, HSS **330**, PGW **332**, and PCRF **334** coupled with one another over interfaces (or "reference points") as shown. The NFs in the EPC **322** are briefly introduced as follows.

[0064] The MME **324** implements mobility management functions to track a current location of the UE **302** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

[0065] The SGW **326** terminates an S1 interface toward the RAN **310** and routes data packets between the RAN **310** and the EPC **322**. The SGW **326** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

[0066] The SGSN **328** tracks a location of the UE **302** and performs security functions and access control. The SGSN **328** also performs inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **324**; MME **324** selection for handovers; etc. The S3 reference point between the MME **324** and the SGSN **328** enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

[0067] The HSS **330** includes a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **330** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **330** and the MME **324** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the EPC **320**.

[0068] The PGW **332** may terminate an SGi interface toward a data network (DN) **336** that may include an application (app)/content server **338**. The PGW **332** routes data packets between the EPC **322** and the data network **336**. The PGW **332** is communicatively coupled with the SGW **326** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **332** may further include a node for policy enforcement and charging data collection (e.g., PCEF). Additionally, the SGi reference point may communicatively couple the PGW **332** with the same or different data network **336**. The PGW **332** may be communicatively coupled with a PCRF **334** via a Gx reference point.

[0069] The PCRF **334** is the policy and charging control element of the EPC **322**. The PCRF **334** is communicatively coupled to the app/content server **338** to determine appropriate QoS and charging parameters for service flows. The PCRF **332** also provisions associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

[0070] The CN **320** may be a 5GC **340** including an AUSF **342**, AMF **344**, SMF **346**, UPF **348**, NSSF **350**, NEF **352**, NRF **354**, PCF **356**, UDM **358**, and AF **360** coupled with one another over various interfaces as shown. The NFs in the 5GC **340** are briefly introduced as follows.

[0071] The AUSF **342** stores data for authentication of UE **302** and handle authentication-related functionality. The AUSF **342** may facilitate a common authentication framework for various access types.

[0072] The AMF **344** allows other functions of the 5GC **340** to communicate with the UE **302** and the RAN **304** and to subscribe to notifications about mobility events with respect to the UE **302**. The AMF **344** is also responsible for registration management (e.g., for registering UE **302**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **344** provides transport for SM messages between the UE **302** and the SMF **346**, and acts as a transparent proxy for routing SM messages. AMF **344** also provides transport for SMS messages between UE **302** and an SMSF. AMF **344** interacts with the AUSF **342** and the UE **302** to perform various security anchor and context management functions. Furthermore, AMF **344** is a termination point of a RAN-CP interface, which includes the N2 reference point between the RAN **304** and the AMF **344**. The AMF **344** is also a termination point of NAS (N1) signaling, and performs NAS ciphering and integrity protection.

[0073] AMF **344** also supports NAS signaling with the UE **302** over an N3IWF interface. The N3IWF provides access to untrusted entities. N3IWF may be a termination point for the N2 interface between the (R)AN **304** and the AMF **344** for the control plane, and may be a termination point for the N3 reference point between the (R)AN **314** and the **348** for the user plane. As such, the AMF **344** handles N2 signalling from the SMF **346** and the AMF **344** for PDU sessions and QoS, encapsulate/de-encapsulate packets for IPSec and N3 tunnelling, marks N3 user-plane packets in the uplink, and enforces QoS corresponding to N3 packet marking taking into account QOS requirements associated with such marking received over N2. N3IWF may also relay UL and DL control-plane NAS signalling between the UE **302** and AMF **344** via an N1 reference point between the UE **302** and the AMF **344**, and relay uplink and downlink user-plane packets between the UE **302** and UPF **348**. The N3IWF also provides mechanisms for IPsec tunnel establishment with the UE **302**. The AMF **344** may exhibit an Namf service-based interface, and may be a termination point for an N14 reference point between two AMFs **344** and an N17 reference point between the AMF **344** and a 5G-EIR (not shown by FIG. **3**).

[0074] The SMF **346** is responsible for SM (e.g., session establishment, tunnel management between UPF **348** and AN **308**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **348** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **344** over N2 to AN **308**; and determining SSC mode of a session. SM refers to management of a PDU session, and a PDU session or "session" refers to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **302** and the DN **336**.

[0075] The UPF **348** acts as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to data network **336**, and a branching point to support multi-homed PDU session. The UPF **348** also performs packet routing and forwarding, packet inspection, enforces user plane part of policy rules, lawfully intercept packets (UP collection), performs traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), performs uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and performs downlink packet buffering and downlink data notification triggering. UPF **348** may include an uplink classifier to support routing traffic flows to a data network.

[0076] The NSSF **350** selects a set of network slice instances serving the UE **302**. The NSSF **350** also determines allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **350** also determines an AMF set to be used to serve the UE **302**, or a list of candidate AMFs **344** based on a suitable configuration and possibly by querying the NRF **354**. The selection of a set of network slice instances for the UE **302** may be triggered by the AMF **344** with which the UE **302** is registered by interacting with the NSSF **350**; this may lead to a change of AMF **344**. The NSSF **350** interacts with the AMF **344** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown).

[0077] The NEF **352** securely exposes services and capabilities provided by 3GPP NFs for third party, internal exposure/re-exposure, AFs **360**, edge computing or fog computing systems (e.g., edge compute node, etc. In such embodiments, the NEF **352** may authenticate, authorize, or throttle the AFs. NEF **352** may also translate information exchanged with the AF **360** and information exchanged with internal network functions. For example, the NEF **352** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **352** may also receive information from other NFs based on exposed capabilities of other NFs. This information may be stored at the NEF **352** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **352** to other NFs and AFs, or used for other purposes such as analytics.

[0078] The NRF **354** supports service discovery functions, receives NF discovery requests from NF instances, and provides information of the discovered NF instances to the requesting NF instances. NRF **354** also maintains information of available NF instances and their supported services. The NRF **354** also supports service discovery functions, wherein the NRF **354** receives NF Discovery Request from NF instance or an SCP (not shown), and provides information of the discovered NF instances to the NF instance or SCP.

[0079] The PCF **356** provides policy rules to control plane functions to enforce them, and may also support unified policy framework to govern network behavior. The PCF **356** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **358**. In addition to communicating with functions over reference points as shown, the PCF **356** exhibit an Npcf service-based interface.

[0080] The UDM **358** handles subscription-related information to support the network entities' handling of communication sessions, and stores subscription data of UE **302**. For example, subscription data may be communicated via an N8 reference point between the UDM **358** and the AMF **344**. The UDM **358** may include two parts, an application front end and a UDR. The UDR may store subscription data and policy data for the UDM **358** and the PCF **356**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **302**) for the NEF **352**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **358**, PCF **356**, and NEF **352** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **358** may exhibit the Nudm service-based interface.

[0081] AF **360** provides application influence on traffic routing, provide access to NEF **352**, and interact with the policy framework for policy control. The AF **360** may influence UPF **348** (*re*) selection and traffic routing. Based on operator deployment, when AF **360** is considered to be a trusted entity, the network operator may permit AF **360** to interact directly with relevant NFs. Additionally, the AF **360** may be used for edge computing implementations,

[0082] The 5GC **340** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **302** is attached to the network. This may reduce latency and load on the network. In edge computing implementations, the 5GC **340** may select a UPF **348** close to the UE **302** and execute traffic steering from the UPF **348** to DN **336** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **360**, which allows the AF **360** to influence UPF (re) selection and traffic routing.

[0083] The data network (DN) **336** may represent various network operator services, Internet access, or third party services that may be provided by one or more servers including, for example, application (app)/content server **338**. The DN **336** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. In this embodiment, the app server **338** can be coupled to an IMS via an S-CSCF or the I-CSCF. In some implementations, the DN **336** may represent one or more local area DNs (LADNs), which are DNs **336** (or DN names (DNNs)) that is/are accessible by a UE **302** in one or more specific areas. Outside of these specific areas, the UE **302** is not able to access the LADN/DN **336**.

[0084] Additionally or alternatively, the DN **336** may be an Edge DN **336**, which is a (local) Data Network that supports the architecture for enabling edge applications. In these embodiments, the app server **338** may represent the physical hardware systems/devices providing app server functionality and/or the application software resident in the cloud or at an edge compute node that performs server function(s). In some embodiments, the app/content server **338** provides an edge hosting environment that provides support required for Edge Application Server's execution.

[0085] In some embodiments, the 5GS can use one or more edge compute nodes to provide an interface and offload processing of wireless communication traffic. In these embodiments, the edge compute nodes may be included in, or co-located with one or more RAN **310**, **314**. For example, the edge compute nodes can provide a connection between the RAN **314** and UPF **348** in the 5GC **340**. The edge compute nodes can use one or more NFV instances instantiated on virtualization infrastructure within the edge compute nodes to process wireless connections to and from the RAN **314** and UPF **348**.

[0086] The interfaces of the 5GC **340** include reference points and service-based itnterfaces. The reference points include: N1 (between the UE **302** and the AMF **344**), N2 (between RAN **314** and AMF **344**), N3 (between RAN **314** and UPF **348**), N4 (between the SMF **346** and UPF **348**), N5 (between PCF **356** and AF **360**), N6 (between UPF **348** and DN **336**), N7 (between SMF **346** and PCF **356**), N8 (between UDM **358** and AMF **344**), N9 (between two UPFs **348**), N10 (between the UDM **358** and the SMF **346**), N11 (between the AMF **344** and the SMF **346**), N12 (between

AUSF **342** and AMF **344**), N13 (between AUSF **342** and UDM **358**), N14 (between two AMFs **344**; not shown), N15 (between PCF **356** and AMF **344** in case of a non-roaming scenario, or between the PCF **356** in a visited network and AMF **344** in case of a roaming scenario), N16 (between two SMFs **346**; not shown), and N22 (between AMF **344** and NSSF **350**). Other reference point representations not shown in FIG. **3** can also be used. The service-based representation of FIG. **3** represents NFs within the control plane that enable other authorized NFs to access their services. The service-based interfaces (SBIs) include: Namf (SBI exhibited by AMF **344**), Nsmf (SBI exhibited by SMF **346**), Nnef (SBI exhibited by NEF **352**), Npcf (SBI exhibited by PCF **356**), Nudm (SBI exhibited by the UDM **358**), Naf (SBI exhibited by AF **360**), Nnrf (SBI exhibited by NRF **354**), Nnssf (SBI exhibited by NSSF **350**), Nausf (SBI exhibited by AUSF **342**). Other service-based interfaces (e.g., Nudr, N5g-eir, and Nudsf) not shown in FIG. **3** can also be used. In some embodiments, the NEF **352** can provide an interface to edge compute nodes **336***x*, which can be used to process wireless connections with the RAN **314**. In some implementations, the system **300** may include an SMSF, which is responsible for SMS subscription checking and verification, and relaying SM messages to/from the UE **302** to/from other entities, such as an SMS-GMSC/IWMSC/SMS-router. The SMS may also interact with AMF **344** and UDM **358** for a notification procedure that the UE **302** is available for SMS transfer (e.g., set a UE not reachable flag, and notifying UDM **358** when UE **302** is available for SMS).

[0087] The 5GS may also include an SCP (or individual instances of the SCP) that supports indirect communication (see e.g., 3GPP TS 23.501 section 7.1.1); delegated discovery (see e.g., 3GPP TS 23.501 section 7.1.1); message forwarding and routing to destination NF/NF service(s), communication security (e.g., authorization of the NF Service Consumer to access the NF Service Producer API) (see e.g., 3GPP TS 33.501), load balancing, monitoring, overload control, etc.; and discovery and selection functionality for UDM(s), AUSF(s), UDR(s), PCF(s) with access to subscription data stored in the UDR based on UE's SUPI, SUCI or GPSI (see e.g., 3GPP TS 23.501 section 6.3). Load balancing, monitoring, overload control functionality provided by the SCP may be implementation specific. The SCP may be deployed in a distributed manner. More than one SCP can be present in the communication path between various NF Services. The SCP, although not an NF instance, can also be deployed distributed, redundant, and scalable.

[0088] FIG. **4** schematically illustrates a wireless network **400** in accordance with various embodiments. The wireless network **400** may include a UE **402** in wireless communication with an AN **404**. The UE **402** and AN **404** may be similar to, and substantially interchangeable with, like-named components described with respect to FIG. **3**.

[0089] The UE **402** may be communicatively coupled with the AN **404** via connection **406**. The connection **406** is illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHZ frequencies.

[0090] The UE **402** may include a host platform **408** coupled with a modem platform **410**. The host platform **408** may include application processing circuitry **412**, which may be coupled with protocol processing circuitry **414** of the modem platform **410**. The application processing circuitry **412** may run various applications for the UE **402** that source/sink application data. The application processing circuitry **412** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example, IP) operations

[0091] The protocol processing circuitry **414** may implement one or more of layer operations to facilitate transmission or reception of data over the connection **406**. The layer operations implemented by the protocol processing circuitry **414** may include, for example, MAC, RLC, PDCP, RRC and NAS operations.

[0092] The modem platform **410** may further include digital baseband circuitry **416** that may implement one or more layer operations that are "below" layer operations performed by the protocol processing circuitry **414** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ acknowledgement (ACK) functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

[0093] The modem platform **410** may further include transmit circuitry **418**, receive circuitry **420**, RF circuitry **422**, and RF front end (RFFE) **424**, which may include or connect to one or more antenna panels **426**. Briefly, the transmit circuitry **418** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **420** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **422** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **424** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **418**, receive circuitry **420**, RF circuitry **422**, RFFE **424**, and antenna panels **426** (referred generically as "transmit/receive components") may be specific to details of a specific implementation such as, for example, whether communication is TDM or FDM, in mmWave or sub-6 gHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed in the same or different chips/modules, etc.

[0094] In some embodiments, the protocol processing circuitry **414** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

[0095] A UE **402** reception may be established by and via the antenna panels **426**, RFFE **424**, RF circuitry **422**, receive circuitry **420**, digital baseband circuitry **416**, and protocol processing circuitry **414**. In some embodiments, the antenna panels **426** may receive a transmission from the AN **404** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **426**.

[0096] A UE **402** transmission may be established by and via the protocol processing circuitry **414**, digital baseband circuitry **416**, transmit circuitry **418**, RF circuitry **422**, RFFE

424, and antenna panels **426**. In some embodiments, the transmit components of the UE **404** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **426**.

[0097] Similar to the UE **402**, the AN **404** may include a host platform **428** coupled with a modem platform **430**. The host platform **428** may include application processing circuitry **432** coupled with protocol processing circuitry **434** of the modem platform **430**. The modem platform may further include digital baseband circuitry **436**, transmit circuitry **438**, receive circuitry **440**, RF circuitry **442**, RFFE circuitry **444**, and antenna panels **446**. The components of the AN **404** may be similar to and substantially interchangeable with like-named components of the UE **402**. In addition to performing data transmission/reception as described above, the components of the AN **408** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

[0098] FIG. **5** illustrates components of a computing device **500** according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **5** shows a diagrammatic representation of hardware resources **501** including one or more processors (or processor cores) **510**, one or more memory/storage devices **520**, and one or more communication resources **530**, each of which may be communicatively coupled via a bus **540** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **502** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **501**.

[0099] The processors **510** include, for example, processor **512** and processor **514**. The processors **510** include circuitry such as, but not limited to one or more processor cores and one or more of cache memory, low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I2C or universal programmable serial interface circuit, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as secure digital/multi-media card (SD/MMC) or similar, interfaces, mobile industry processor interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors **510** may be, for example, a central processing unit (CPU), reduced instruction set computing (RISC) processors, Acorn RISC Machine (ARM) processors, complex instruction set computing (CISC) processors, graphics processing units (GPUs), one or more Digital Signal Processors (DSPs) such as a baseband processor, Application-Specific Integrated Circuits (ASICs), an Field-Programmable Gate Array (FPGA), a radio-frequency integrated circuit (RFIC), one or more microprocessors or controllers, another processor (including those discussed herein), or any suitable combination thereof. In some implementations, the processor circuitry **510** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices (e.g., FPGA, complex programmable logic devices (CPLDs), etc.), or the like.

[0100] The memory/storage devices **520** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **520** may include, but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as random access memory (RAM), dynamic RAM (DRAM), static RAM (SRAM), synchronous DRAM (SDRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, phase change RAM (PRAM), resistive memory such as magnetoresistive random access memory (MRAM), etc., and may incorporate three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®. The memory/storage devices **520** may also comprise persistent storage devices, which may be temporal and/or persistent storage of any type, including, but not limited to, non-volatile memory, optical, magnetic, and/or solid state mass storage, and so forth.

[0101] The communication resources **530** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **504** or one or more databases **506** or other network elements via a network **508**. For example, the communication resources **530** may include wired communication components (e.g., for coupling via USB, Ethernet, Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), Ethernet over USB, Controller Area Network (CAN), Local Interconnect Network (LIN), DeviceNet, ControlNet, Data Highway+, PROFIBUS, or PROFINET, among many others), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, WiFi® components, and other communication components. Network connectivity may be provided to/from the computing device **500** via the communication resources **530** using a physical connection, which may be electrical (e.g., a "copper interconnect") or optical. The physical connection also includes suitable input connectors (e.g., ports, receptacles, sockets, etc.) and output connectors (e.g., plugs, pins, etc.). The communication resources **530** may include one or more dedicated processors and/or FPGAs to communicate using one or more of the aforementioned network interface protocols.

[0102] Instructions **550** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **510** to perform any one or more of the methodologies discussed herein. The instructions **550** may reside, completely or partially, within at least one of the processors **510** (e.g., within the processor's cache memory), the memory/storage devices **520**, or any suitable combination thereof. Furthermore, any portion of the instructions **550** may be transferred to the hardware resources **501** from any combination of the peripheral devices **504** or the databases **506**. Accordingly, the memory of processors **510**, the memory/storage devices **520**, the peripheral devices **504**, and the databases **506** are examples of computer-readable and machine-readable media.

[0103] FIG. **6** illustrates a network **600** in accordance with various embodiments. The network **600** may operate in a matter consistent with 3GPP technical specifications or technical reports for 6G systems. In some embodiments, the network **600** may operate concurrently with network **300**. For example, in some embodiments, the network **600** may share one or more frequency or bandwidth resources with network **300**. As one specific example, a UE (e.g., UE **602**) may be configured to operate in both network **600** and network **300**. Such configuration may be based on a UE including circuitry configured for communication with frequency and bandwidth resources of both networks **300** and **600**. In general, several elements of network **600** may share one or more characteristics with elements of network **300**. For the sake of brevity and clarity, such elements may not be repeated in the description of network **600**.

[0104] The network **600** may include a UE **602**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **608** via an over-the-air connection. The UE **602** may be similar to, for example, UE **302**. The UE **602** may be, but is not limited to, a smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

[0105] Although not specifically shown in FIG. **6**, in some embodiments the network **600** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc. Similarly, although not specifically shown in FIG. **6**, the UE **602** may be communicatively coupled with an AP such as AP **306** as described with respect to FIG. **3**. Additionally, although not specifically shown in FIG. **6**, in some embodiments the RAN **608** may include one or more ANss such as AN **308** as described with respect to FIG. **3**. The RAN **608** and/or the AN of the RAN **608** may be referred to as a base station (BS), a RAN node, or using some other term or name.

[0106] The UE **602** and the RAN **608** may be configured to communicate via an air interface that may be referred to as a sixth generation (6G) air interface. The 6G air interface may include one or more features such as communication in a terahertz (THz) or sub-THz bandwidth, or joint communication and sensing. As used herein, the term "joint communication and sensing" may refer to a system that allows for wireless communication as well as radar-based sensing via various types of multiplexing. As used herein, THz or sub-THz bandwidths may refer to communication in the 80 GHz and above frequency ranges. Such frequency ranges may additionally or alternatively be referred to as "millimeter wave" or "mmWave" frequency ranges.

[0107] The RAN **608** may allow for communication between the UE **602** and a 6G core network (CN) **610**. Specifically, the RAN **608** may facilitate the transmission and reception of data between the UE **602** and the 6G CN **610**. The 6G CN **610** may include various functions such as NSSF **350**, NEF **352**, NRF **354**, PCF **356**, UDM **358**, AF **360**, SMF **346**, and AUSF **342**. The 6G CN **610** may additional include UPF **348** and DN **336** as shown in FIG. **6**.

[0108] Additionally, the RAN **608** may include various additional functions that are in addition to, or alternative to, functions of a legacy cellular network such as a 4G or 5G network. Two such functions may include a Compute Control Function (Comp CF) **624** and a Compute Service Function (Comp SF) **636**. The Comp CF **624** and the Comp SF **636** may be parts or functions of the Computing Service Plane. Comp CF **624** may be a control plane function that provides functionalities such as management of the Comp SF **636**, computing task context generation and management (e.g., create, read, modify, delete), interaction with the underlaying computing infrastructure for computing resource management, etc., Comp SF **636** may be a user plane function that serves as the gateway to interface computing service users (such as UE **602**) and computing nodes behind a Comp SF instance. Some functionalities of the Comp SF **636** may include: parse computing service data received from users to compute tasks executable by computing nodes; hold service mesh ingress gateway or service API gateway; service and charging policies enforcement; performance monitoring and telemetry collection, etc. In some embodiments, a Comp SF **636** instance may serve as the user plane gateway for a cluster of computing nodes. A Comp CF **624** instance may control one or more Comp SF **636** instances.

[0109] Two other such functions may include a Communication Control Function (Comm CF) **628** and a Communication Service Function (Comm SF) **638**, which may be parts of the Communication Service Plane. The Comm CF **628** may be the control plane function for managing the Comm SF **638**, communication sessions creation/configuration/releasing, and managing communication session context. The Comm SF **638** may be a user plane function for data transport. Comm CF **628** and Comm SF **638** may be considered as upgrades of SMF **346** and UPF **348**, which were described with respect to a 5G system in FIG. **3**. The upgrades provided by the Comm CF **628** and the Comm SF **638** may enable service-aware transport. For legacy (e.g., 4G or 5G) data transport, SMF **346** and UPF **348** may still be used.

[0110] Two other such functions may include a Data Control Function (Data CF) **622** and Data Service Function (Data SF) **632** may be parts of the Data Service Plane. Data CF **622** may be a control plane function and provides functionalities such as Data SF **632** management, Data service creation/configuration/releasing, Data service context management, etc. Data SF **632** may be a user plane function and serve as the gateway between data service users (such as UE **602** and the various functions of the 6G CN **610**) and data service endpoints behind the gateway. Specific functionalities may include include: parse data service user data and forward to corresponding data service endpoints, generate charging data, report data service status.

[0111] Another such function may be the Service Orchestration and Chaining Function (SOCF) **620**, which may discover, orchestrate and chain up communication/computing/data services provided by functions in the network. Upon receiving service requests from users, SOCF **620** may interact with one or more of Comp CF **624**, Comm CF **628**, and Data CF **622** to identify Comp SF **636**, Comm SF **638**, and Data SF **632** instances, configure service resources, and generate the service chain, which could contain multiple Comp SF **636**, Comm SF **638**, and Data SF **632** instances and their associated computing endpoints. Workload processing and data movement may then be conducted within the generated service chain. The SOCF **620** may also responsible for maintaining, updating, and releasing a created service chain.

[0112] Another such function may be the service registration function (SRF) **614**, which may act as a registry for system services provided in the user plane such as services provided by service endpoints behind Comp SF **636** and

Data SF **632** gateways and services provided by the UE **602**. The SRF **614** may be considered a counterpart of NRF **354**, which may act as the registry for network functions.

[0113] Other such functions may include an evolved service communication proxy (eSCP) and service infrastructure control function (SICF) **626**, which may provide service communication infrastructure for control plane services and user plane services. The eSCP may be related to the service communication proxy (SCP) of 5G with user plane service communication proxy capabilities being added. The eSCP is therefore expressed in two parts: eCSP-C **612** and eSCP-U **634**, for control plane service communication proxy and user plane service communication proxy, respectively. The SICF **626** may control and configure eCSP instances in terms of service traffic routing policies, access rules, load balancing configurations, performance monitoring, etc.

[0114] Another such function is the AMF **644**. The AMF **644** may be similar to **344**, but with additional functionality. Specifically, the AMF **644** may include potential functional repartition, such as move the message forwarding functionality from the AMF **644** to the RAN **608**.

[0115] Another such function is the service orchestration exposure function (SOEF) **618**. The SOEF may be configured to expose service orchestration and chaining services to external users such as applications.

[0116] The UE **602** may include an additional function that is referred to as a computing client service function (comp CSF) **604**. The comp CSF **604** may have both the control plane functionalities and user plane functionalities, and may interact with corresponding network side functions such as SOCF **620**, Comp CF **624**, Comp SF **636**, Data CF **622**, and/or Data SF **632** for service discovery, request/response, compute task workload exchange, etc. The Comp CSF **604** may also work with network side functions to decide on whether a computing task should be run on the UE **602**, the RAN **608**, and/or an element of the 6G CN **610**.

[0117] The UE **602** and/or the Comp CSF **604** may include a service mesh proxy **606**. The service mesh proxy **606** may act as a proxy for service-to-service communication in the user plane. Capabilities of the service mesh proxy **606** may include one or more of addressing, security, load balancing, etc.

[0118] FIG. **7** illustrates a simplified block diagram of artificial (AI)-assisted communication between a UE **705** and a RAN **710**, in accordance with various embodiments. More specifically, as described in further detail below, AI/machine learning (ML) models may be used or leveraged to facilitate over-the-air communication between UE **705** and RAN **710**.

[0119] One or both of the UE **705** and the RAN **710** may operate in a matter consistent with 3GPP technical specifications or technical reports for 6G systems. In some embodiments, the wireless cellular communication between the UE **705** and the RAN **710** may be part of, or operate concurrently with, networks **600**, **300**, and/or some other network described herein.

[0120] The UE **705** may be similar to, and share one or more features with, UE **602**, UE **302**, and/or some other UE described herein. The UE **705** may be, but is not limited to, a smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc. The RAN **710** may be similar to, and share one or more features with, RAN **314**, RAN **608**, and/or some other RAN described herein.

[0121] As may be seen in FIG. **7**, the AI-related elements of UE **705** may be similar to the AI-related elements of RAN **710**. For the sake of discussion herein, description of the various elements will be provided from the point of view of the UE **705**, however it will be understood that such discussion or description will apply to equally named/numbered elements of RAN **710**, unless explicitly stated otherwise.

[0122] As previously noted, the UE **705** may include various elements or functions that are related to AI/ML. Such elements may be implemented as hardware, software, firmware, and/or some combination thereof. In embodiments, one or more of the elements may be implemented as part of the same hardware (e.g., chip or multi-processor chip), software (e.g., a computing program), or firmware as another element.

[0123] One such element may be a data repository **715**. The data repository **715** may be responsible for data collection and storage. Specifically, the data repository **715** may collect and store RAN configuration parameters, measurement data, performance key performance indicators (KPIs), model performance metrics, etc., for model training, update, and inference. More generally, collected data is stored into the repository. Stored data can be discovered and extracted by other elements from the data repository **715**. For example, as may be seen, the inference data selection/filter element **750** may retrieve data from the data repository **715**. In various embodiments, the UE **705** may be configured to discover and request data from the data repository **715** in the RAN, and vice versa. More generally, the data repository **715** of the UE **705** may be communicatively coupled with the data repository **715** of the RAN **710** such that the respective data repositories of the UE and the RAN may share collected data with one another.

[0124] Another such element may be a training data selection/filtering functional block **720**. The training data selection/filter functional block **720** may be configured to generate training, validation, and testing datasets for model training. Training data may be extracted from the data repository **715**. Data may be selected/filtered based on the specific AI/ML model to be trained. Data may optionally be transformed/augmented/pre-processed (e.g., normalized) before being loaded into datasets. The training data selection/filter functional block **720** may label data in datasets for supervised learning. The produced datasets may then be fed into model training the model training functional block **725**.

[0125] As noted above, another such element may be the model training functional block **725**. This functional block may be responsible for training and updating (re-training) AI/ML models. The selected model may be trained using the fed-in datasets (including training, validation, testing) from the training data selection/filtering functional block. The model training functional block **725** may produce trained and tested AI/ML models which are ready for deployment. The produced trained and tested models can be stored in a model repository **735**.

[0126] The model repository **735** may be responsible for AI/ML models' (both trained and un-trained) storage and exposure. Trained/updated model(s) may be stored into the model repository **735**. Model and model parameters may be discovered and requested by other functional blocks (e.g., the training data selection/filter functional block **720** and/or the model training functional block **725**). In some embodiments, the UE **705** may discover and request AI/ML models from the model repository **735** of the RAN **710**. Similarly, the RAN **710** may be able to discover and/or request AI/ML models from the model repository **735** of the UE **705**. In some embodiments, the RAN **710** may configure models and/or model parameters in the model repository **735** of the UE **705**.

[0127] Another such element may be a model management functional block **740**. The model management functional block **740** may be responsible for management of the AI/ML model produced by the model training functional block **725**. Such management functions may include deployment of a trained model, monitoring model performance, etc. In model deployment, the model management functional block **740** may allocate and schedule hardware and/or software resources for inference, based on received trained and tested models. As used herein, "inference" refers to the process of using trained AI/ML model(s) to generate data analytics, actions, policies, etc. based on input inference data. In performance monitoring, based on wireless performance KPIs and model performance metrics, the model management functional block **740** may decide to terminate the running model, start model re-training, select another model, etc. In embodiments, the model management functional block **740** of the RAN **710** may be able to configure model management policies in the UE **705** as shown.

[0128] Another such element may be an inference data selection/filtering functional block **750**. The inference data selection/filter functional block **750** may be responsible for generating datasets for model inference at the inference functional block **745**, as described below. Specifically, inference data may be extracted from the data repository **715**. The inference data selection/filter functional block **750** may select and/or filter the data based on the deployed AI/ML model. Data may be transformed/augmented/pre-processed following the same transformation/augmentation/pre-processing as those in training data selection/filtering as described with respect to functional block **720**. The produced inference dataset may be fed into the inference functional block **745**.

[0129] Another such element may be the inference functional block **745**. The inference functional block **745** may be responsible for executing inference as described above. Specifically, the inference functional block **745** may consume the inference dataset provided by the inference data selection/filtering functional block **750**, and generate one or more outcomes. Such outcomes may be or include data analytics, actions, policies, etc. The outcome(s) may be provided to the performance measurement functional block **730**.

[0130] The performance measurement functional block **730** may be configured to measure model performance metrics (e.g., accuracy, model bias, run-time latency, etc.) of deployed and executing models based on the inference outcome(s) for monitoring purpose. Model performance data may be stored in the data repository **715**.

[0131] For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

[0132] Additional examples of the presently described embodiments include the following, non-limiting implementations. Each of the following non-limiting examples may stand on its own or may be combined in any permutation or combination with any one or more of the other examples provided below or throughout the present disclosure.

[0133] For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below.

[0134] The following examples pertain to further embodiments.

[0135] Example 1 may include the method for supporting AI/ML based positioning in 5GS using LMF performing the inference for AI/ML based positioning method and NWDAF containing MTLF performs the ML model training for UE AI/ML positioning analytics in the 5GC.

[0136] Example 2 may include the method of example 1 and/or some other example herein, where the LMF decision to get a trained ML model from NWDAF may be based on LMF implementation or based on ML model accuracy monitoring requirement by the service consumer of the trained ML model for UE AIML positioning analytics.

[0137] Example 3 may include the method of example 2 and/or some other example herein, where the LMF in the role of service consumer sends the Nnwdaf_MLModelTraining_Subscribe for UE AIML positioning analytics.

[0138] Example 4 may include the method of example 3 and/or some other example herein, where the NWDAF MTLF initiates the data collection required to training train the ML model for UE AIML positioning analytics from the GMLC using Ngmlc_Location_ProvideLocationMeasurement request for a UE or group of UEs which may be determined based on the ML Model Filter Information.

[0139] Example 5 may include the method of example 4 and/or some other example herein, where the GMLC sends Namf_Location_ProvideAIMLPosMeasurement Request to the AMF(s) serving the UE or group of UEs.

[0140] Example 6 may include the method of example 5 and/or some other example herein, where the AMF sends the Nlmf_Location_MeasurementDataRequest to the LMF(s) with the AIMLpositioning indication to indicate that the measurement data request is for AIML positioning method.

[0141] Example 7 may include the method of example 6 and/or some other example herein, where the LMF sends the Nlmf_Location_MeasurementData response to the AMF with the requested location measurement data or the ADRF ID plus DataSetTag where the AIML based measurement data is stored.

[0142] Example 8 may include the method of example 7 and/or some other example herein, where the LMF may send the ADRF ID plus DataSetTag if the LMF stores the measurement data collected during ML model inference, the LMF stores the measurement data collected during ML model inference in the ADRF, then the ADRF ID and DataSetTag can be sent to the consumer (NWDAF MTLF) to retrieve the input data required for ML model training.

[0143] Example 9 may include the method of example 8 and/or some other example herein, where if the requested measurement data requested is not available at the LMF or ADRF, the LMF will collect different measurement data from UE or RAN for ML model training to obtain UE location and positioning measurement data for UE AI/ML positioning analytics.

[0144] Example 10 may include the method of example 8, 9, and/or some other example herein, where the AMF sends the Namf_Location_ProvideAIMLPosMeasurement Response to the GMLC with the location measurement data or the ADRF ID plus DataSetTag where the AIML based measurement data is stored.

[0145] Example 11 may include the method of example 10 and/or some other example herein, where GMLC may aggregate one or more UE location measurement data received or ADRF ID plus DataSetTag reports in each message sent to NWDAF containing MTLF.

[0146] Example 12 may include the method of example 11 and/or some other example herein, where the NWDAF containing MTLF sends the Nnwdaf_MLModelTraining_Notify for AIML positioning analytics to the LMF.

[0147] Example 13 may include a method to be performed by an AMF, one or more elements of an AMF, and/or one or more electronic devices that include and/or implement an AMF, wherein the method comprises: identifying a location management function (LMF) to be associated with one or more user equipments (UEs); transmitting, to the LMF, a measurement data request associated with artificial intelligence/machine learning (AI/ML) positioning of the one or more UEs; identifying, from the LMF based on the measurement data request, a response that includes an indication of the measurement data; transmitting an indication of the measurement data to a gateway mobile location centre (GMLC); and identifying, based on the transmission of the indication of the measurement data to the GMLC, an indication of an AI/ML model to be used for positioning of the one or more UEs.

[0148] Example 14 may include the method of example 13, and/or some other example herein, wherein the measurement data request is a Nlmf_Location_Measurement-DataRequest.

[0149] Example 15 may include the method of any of examples 13-14, and/or some other example herein, wherein the transmission of the measurement data request is based on an identification that one or more positioning measurement parameters to be collected for the AI/ML positioning are different from one or more positioning measurement parameters of a radio access network (RAN) with which the AMF is associated.

[0150] Example 16 may include the method of any of examples 13-15, and/or some other example herein, wherein the response is a Nlmf_Location_MeasurementData response.

[0151] Example 17 may include the method of any of examples 13-16, and/or some other example herein, wherein the indication of the measurement data includes the measurement data.

[0152] Example 18 may include the method of any of examples 13-17, and/or some other example herein, wherein the indication of the measurement data includes a pointer to a location where the measurement data is stored.

[0153] Example 19 may include the method of any of examples 13-18, and/or some other example herein, wherein the AI/ML model is trained based on the measurement data.

[0154] Example 37 may include an apparatus comprising means for performing any of the methods of examples 1-36.

[0155] Example 38 may include a network node comprising a communication interface and processing circuitry connected thereto and configured to perform the methods of examples 1-36.

[0156] Example 39 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

[0157] Example 40 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

[0158] Example 41 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

[0159] Example 42 may include a method, technique, or process as described in or related to any of examples 1-36, or portions or parts thereof.

[0160] Example 43 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-36, or portions thereof.

[0161] Example 44 may include a signal as described in or related to any of examples 1-36, or portions or parts thereof.

[0162] Example 45 may include a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-36, or portions or parts thereof, or otherwise described in the present disclosure.

[0163] Example 46 may include a signal encoded with data as described in or related to any of examples 1-36, or portions or parts thereof, or otherwise described in the present disclosure.

[0164] Example 47 may include a signal encoded with a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-36, or portions or parts thereof, or otherwise described in the present disclosure.

[0165] Example 48 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-36, or portions thereof.

[0166] Example 49 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples 1-36, or portions thereof.

[0167] Example 50 may include a signal in a wireless network as shown and described herein.

[0168] Example 51 may include a method of communicating in a wireless network as shown and described herein.

[0169] Example 52 may include a system for providing wireless communication as shown and described herein.

[0170] Example 53 may include a device for providing wireless communication as shown and described herein.

[0171] An example implementation is an edge computing system, including respective edge processing devices and nodes to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is a client endpoint node, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an aggregation node, network hub node, gateway node, or core data processing node, within or coupled to an edge computing system, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an access point, base station, road-side unit, street-side unit, or on-premise unit, within or coupled to an edge computing system, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an edge provisioning node, service orchestration node, application orchestration node, or multi-tenant management node, within or coupled to an edge computing system, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an edge node operating an edge provisioning service, application or service orchestration service, virtual machine deployment, container deployment, function deployment, and compute management, within or coupled to an edge computing system, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an edge computing system operable as an edge mesh, as an edge mesh with side car loading, or with mesh-to-mesh communications, operable to invoke or perform the operations of the examples above, or other subject matter described herein. Another example implementation is an edge computing system including aspects of network functions, acceleration functions, acceleration hardware, storage hardware, or computation hardware resources, operable to invoke or perform the use cases discussed herein, with use of the examples above, or other subject matter described herein. Another example implementation is an edge computing system adapted for supporting client mobility, vehicle-to-vehicle (V2V), vehicle-to-everything (V2X), or vehicle-to-infrastructure (V2I) scenarios, and optionally operating according to ETSI MEC specifications, operable to invoke or perform the use cases discussed herein, with use of the examples above, or other subject matter described herein. Another example implementation is an edge computing system adapted for mobile wireless communications, including configurations according to an 3GPP 4G/LTE or 5G network capabilities, operable to invoke or perform the use cases discussed herein, with use of the examples above, or other subject matter described herein. Another example implementation is a computing system adapted for network communications, including configurations according to an O-RAN capabilities, operable to invoke or perform the use cases discussed herein, with use of the examples above, or other subject matter described herein.

[0172] Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Terminology

[0173] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a," "an" and "the" are intended to include plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specific the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operation, elements, components, and/or groups thereof.

[0174] For the purposes of the present disclosure, the phrase "A and/or B" means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase "A, B, and/or C" means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B and C). The description may use the phrases "in an embodiment," or "In some embodiments," which may each refer to one or more of the same or different embodiments. Furthermore, the terms "comprising," "including," "having," and the like, as used with respect to embodiments of the present disclosure, are synonymous.

[0175] The terms "coupled," "communicatively coupled," along with derivatives thereof are used herein. The term "coupled" may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term "directly coupled" may mean that two or more elements are in direct contact with one another. The term "communicatively coupled" may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or ink, and/or the like.

[0176] The term "circuitry" as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term "circuitry" may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

[0177] The term "processor circuitry" as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. Processing circuitry may include one or more processing cores to execute instructions and one or more memory structures to store program and data information. The term "processor circuitry" may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. Processing circuitry may include more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. The terms "application circuitry" and/or "baseband circuitry" may be considered synonymous to, and may be referred to as, "processor circuitry."

[0178] The term "memory" and/or "memory circuitry" as used herein refers to one or more hardware devices for storing data, including RAM, MRAM, PRAM, DRAM, and/or SDRAM, core memory, ROM, magnetic disk storage mediums, optical storage mediums, flash memory devices or other machine readable mediums for storing data. The term "computer-readable medium" may include, but is not limited to, memory, portable or fixed storage devices, optical storage devices, and various other mediums capable of storing, containing or carrying instructions or data.

[0179] The term "interface circuitry" as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term "interface circuitry" may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

[0180] The term "user equipment" or "UE" as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term "user equipment" or "UE" may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term "user equipment" or "UE" may include any type of wireless/wired device or any computing device including a wireless communications interface.

[0181] The term "network element" as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term "network element" may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

[0182] The term "computer system" as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term "computer system" and/or "system" may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term "computer system" and/or "system" may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

[0183] The term "appliance," "computer appliance," or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A "virtual appliance" is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource. The term "element" refers to a unit that is indivisible at a given level of abstraction and has a clearly defined boundary, wherein an element may be any type of entity including, for example, one or more devices, systems, controllers, network elements, modules, etc., or combinations thereof. The term "device" refers to a physical entity embedded inside, or attached to, another physical entity in its vicinity, with capabilities to convey digital information from or to that physical entity. The term "entity" refers to a distinct component of an architecture or device, or information transferred as a payload. The term "controller" refers to an element or entity that has the capability to affect a physical entity, such as by changing its state or causing the physical entity to move.

[0184] The term "cloud computing" or "cloud" refers to a paradigm for enabling network access to a scalable and elastic pool of shareable computing resources with self-service provisioning and administration on-demand and without active management by users. Cloud computing provides cloud computing services (or cloud services), which are one or more capabilities offered via cloud computing that are invoked using a defined interface (e.g., an API or the like). The term "computing resource" or simply "resource" refers to any physical or virtual component, or usage of such components, of limited availability within a computer system or network. Examples of computing resources include usage/access to, for a period of time, servers, processor(s), storage equipment, memory devices, memory areas, networks, electrical power, input/output (peripheral) devices, mechanical devices, network connections (e.g., channels/links, ports, network sockets, etc.), operating systems, virtual machines (VMs), software/applications, computer files, and/or the like. A "hardware resource" may refer to compute, storage, and/or network resources provided by physical hardware element(s). A "virtualized resource" may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term "network resource" or "communication resource" may refer to resources that are accessible by computer devices/systems via a communications network. The term "system resources" may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable. As used herein, the term "cloud service provider" (or CSP) indicates an organization which operates typically large-scale "cloud" resources comprised of centralized, regional, and edge data centers (e.g., as used in the context of the public cloud). In other examples, a CSP may also be referred to as a Cloud Service Operator (CSO). References to "cloud computing" generally refer to computing resources and services offered by a CSP or a CSO, at remote locations with at least some increased latency, distance, or constraints relative to edge computing.

[0185] As used herein, the term "data center" refers to a purpose-designed structure that is intended to house multiple high-performance compute and data storage nodes such that a large amount of compute, data storage and network resources are present at a single location. This often entails specialized rack and enclosure systems, suitable heating, cooling, ventilation, security, fire suppression, and power delivery systems. The term may also refer to a compute and data storage node in some contexts. A data center may vary in scale between a centralized or cloud data center (e.g., largest), regional data center, and edge data center (e.g., smallest).

[0186] As used herein, the term "edge computing" refers to the implementation, coordination, and use of computing and resources at locations closer to the "edge" or collection of "edges" of a network. Deploying computing resources at the network's edge may reduce application and network latency, reduce network backhaul traffic and associated energy consumption, improve service capabilities, improve compliance with security or data privacy requirements (especially as compared to conventional cloud computing), and improve total cost of ownership). As used herein, the term "edge compute node" refers to a real-world, logical, or virtualized implementation of a compute-capable element in the form of a device, gateway, bridge, system or subsystem, component, whether operating in a server, client, endpoint, or peer mode, and whether located at an "edge" of an network or at a connected location further within the network. References to a "node" used herein are generally interchangeable with a "device", "component", and "sub-system"; however, references to an "edge computing system" or "edge computing network" generally refer to a distributed architecture, organization, or collection of multiple nodes and devices, and which is organized to accomplish or offer some aspect of services or resources in an edge computing setting.

[0187] Additionally or alternatively, the term "Edge Computing" refers to a concept, as described in [**6**], that enables operator and 3rd party services to be hosted close to the UE's access point of attachment, to achieve an efficient service delivery through the reduced end-to-end latency and load on the transport network. As used herein, the term "Edge Computing Service Provider" refers to a mobile network operator or a 3rd party service provider offering Edge Computing service. As used herein, the term "Edge Data Network" refers to a local Data Network (DN) that supports the architecture for enabling edge applications. As used herein, the term "Edge Hosting Environment" refers to an environment providing support required for Edge Application Server's execution. As used herein, the term "Application Server" refers to application software resident in the cloud performing the server function.

[0188] The term "Internet of Things" or "IoT" refers to a system of interrelated computing devices, mechanical and digital machines capable of transferring data with little or no human interaction, and may involve technologies such as real-time analytics, machine learning and/or AI, embedded systems, wireless sensor networks, control systems, automation (e.g., smarthome, smart building and/or smart city technologies), and the like. IoT devices are usually low-power devices without heavy compute or storage capabilities. "Edge IoT devices" may be any kind of IoT devices deployed at a network's edge.

[0189] As used herein, the term "cluster" refers to a set or grouping of entities as part of an edge computing system (or systems), in the form of physical entities (e.g., different computing systems, networks or network groups), logical entities (e.g., applications, functions, security constructs, containers), and the like. In some locations, a "cluster" is also referred to as a "group" or a "domain". The membership of cluster may be modified or affected based on conditions or functions, including from dynamic or property-based membership, from network or system management scenarios, or from various example techniques discussed below which may add, modify, or remove an entity in a cluster. Clusters may also include or be associated with multiple layers, levels, or properties, including variations in security features and results based on such layers, levels, or properties.

[0190] The term "application" may refer to a complete and deployable package, environment to achieve a certain function in an operational environment. The term "AI/ML application" or the like may be an application that contains some AI/ML models and application-level descriptions. The term "machine learning" or "ML" refers to the use of computer systems implementing algorithms and/or statistical models to perform specific task(s) without using explicit instructions, but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as "ML models" or the like) based on sample data (referred to as "training data," "model training information," or the like) in order to make predictions or decisions without being explicitly programmed to perform such tasks. Generally, an ML algorithm is a computer program that learns from experience with respect to some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term "ML algorithm" refers to different concepts than the term "ML model," these terms as discussed herein may be used interchangeably for the purposes of the present disclosure.

[0191] The term "machine learning model," "ML model," or the like may also refer to ML methods and concepts used by an ML-assisted solution. An "ML-assisted solution" is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), decision tree algorithms, support machine vectors, Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principle component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference. An "ML pipeline" is a set of functionalities, functions, or functional entities specific for an ML-assisted solution; an ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The "actor" is an entity that hosts an ML assisted solution using the output of the ML model inference). The term "ML training host" refers to an entity, such as a network function, that hosts the training of the model. The term "ML inference host" refers to an entity, such as a network function, that hosts model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML-host informs the actor about the output of the ML algorithm, and the actor takes a decision for an action (an "action" is performed by an actor as a result of the output of an ML assisted solution). The term "model inference information" refers to information used as an input to the ML model for determining inference(s); the data used to train an ML model and the data used to determine inferences may overlap, however, "training data" and "inference data" refer to different concepts.

[0192] The terms "instantiate," "instantiation," and the like as used herein refers to the creation of an instance. An "instance" also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code. The term "information element" refers to a structural element containing one or more fields. The term "field" refers to individual contents of an information element, or a data element that contains content. As used herein, a "database object", "data structure", or the like may refer to any representation of information that is in the form of an object, attribute-value pair (AVP), key-value pair (KVP), tuple, etc., and may include variables, data structures, functions, methods, classes, database records, database fields, database entities, associations between data and/or database entities (also referred to as a "relation"), blocks and links between blocks in block chain implementations, and/or the like.

[0193] An "information object," as used herein, refers to a collection of structured data and/or any representation of information, and may include, for example electronic documents (or "documents"), database objects, data structures, files, audio data, video data, raw data, archive files, application packages, and/or any other like representation of information. The terms "electronic document" or "document," may refer to a data structure, computer file, or resource used to record data, and includes various file types and/or data formats such as word processing documents, spreadsheets, slide presentations, multimedia items, webpage and/or source code documents, and/or the like. As examples, the information objects may include markup and/or source code documents such as HTML, XML, JSON, Apex®, CSS, JSP, MessagePack™, Apache® Thrift™, ASN.1, Google® Protocol Buffers (protobuf), or some other document(s)/format(s) such as those discussed herein. An information object may have both a logical and a physical structure. Physically, an information object comprises one or more units called entities. An entity is a unit of storage that contains content and is identified by a name. An entity may refer to other entities to cause their inclusion in the information object. An information object begins in a document entity, which is also referred to as a root element (or "root"). Logically, an information object comprises one or more declarations, elements, comments, character references, and processing instructions, all of which are indicated in the information object (e.g., using markup).

[0194] The term "data item" as used herein refers to an atomic state of a particular object with at least one specific property at a certain point in time. Such an object is usually identified by an object name or object identifier, and properties of such an object are usually defined as database objects (e.g., fields, records, etc.), object instances, or data elements (e.g., mark-up language elements/tags, etc.). Additionally or alternatively, the term "data item" as used herein may refer to data elements and/or content items, although these terms may refer to difference concepts. The term "data element" or "element" as used herein refers to a unit that is indivisible at a given level of abstraction and has a clearly defined boundary. A data element is a logical component of an information object (e.g., electronic document) that may begin with a start tag (e.g., "<element>") and end with a matching end tag (e.g., "</element>"), or only has an empty element tag (e.g., "<element/>"). Any characters between the start tag and end tag, if any, are the element's content (referred to herein as "content items" or the like).

[0195] The content of an entity may include one or more content items, each of which has an associated datatype representation. A content item may include, for example, attribute values, character values, URIs, qualified names (qnames), parameters, and the like. A qname is a fully qualified name of an element, attribute, or identifier in an information object. A qname associates a URI of a namespace with a local name of an element, attribute, or identifier in that namespace. To make this association, the qname assigns a prefix to the local name that corresponds to its namespace. The qname comprises a URI of the namespace, the prefix, and the local name. Namespaces are used to provide uniquely named elements and attributes in information objects. Content items may include text content (e.g., "<element>content item</element>"), attributes (e.g., "<element attribute=" attribute Value ">"), and other elements referred to as "child elements" (e.g., "<element1><element2>content item</element2></element1>"). An "attribute" may refer to a markup construct including a name-value pair that exists within a start tag or empty element tag. Attributes contain data related to its element and/or control the element's behavior.

[0196] The term "resource" as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A "hardware resource" may refer to compute, storage, and/or network resources provided by physical hardware element(s). A "virtualized resource" may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term "network resource" or "communication resource" may refer to resources that are accessible by computer devices/systems via a communications network. The term "system resources" may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable. The term "channel" as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term "channel" may be synonymous with and/or equivalent to "communications channel," "data communications channel," "transmission channel," "data transmission channel," "access channel," "data access channel," "link," "data link," "carrier," "radiofrequency carrier," and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term "link" as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information. As used herein, the term "radio technology" refers to technology for wireless transmission and/or reception of electromagnetic radiation for information transfer. The term "radio access technology" or "RAT" refers to the technology used for the underlying physical connection to a radio based communication network. As used herein, the term "communication protocol" (either wired or wireless) refers to a set of standardized rules or instructions implemented by a communication device and/or system to communicate with other devices and/or systems, including instructions for packetizing/depacketizing data, modulating/demodulating signals, implementation of protocols stacks, and/or the like.

[0197] As used herein, the term "radio technology" refers to technology for wireless transmission and/or reception of electromagnetic radiation for information transfer. The term "radio access technology" or "RAT" refers to the technology used for the underlying physical connection to a radio based communication network. As used herein, the term "communication protocol" (either wired or wireless) refers to a set of standardized rules or instructions implemented by a communication device and/or system to communicate with other devices and/or systems, including instructions for packetizing/depacketizing data, modulating/demodulating signals, implementation of protocols stacks, and/or the like. Examples of wireless communications protocols may be used in various embodiments include a Global System for Mobile Communications (GSM) radio communication technology, a General Packet Radio Service (GPRS) radio communication technology, an Enhanced Data Rates for GSM Evolution (EDGE) radio communication technology, and/or a Third Generation Partnership Project (3GPP) radio communication technology including, for example, 3GPP Fifth Generation (5G) or New Radio (NR), Universal Mobile Telecommunications System (UMTS), Freedom of Multimedia Access (FOMA), Long Term Evolution (LTE), LTE-Advanced (LTE Advanced), LTE Extra, LTE-A Pro, cdmaOne (2G), Code Division Multiple Access 2000 (CDMA 2000), Cellular Digital Packet Data (CDPD), Mobitex, Circuit Switched Data (CSD), High-Speed CSD (HSCSD), Universal Mobile Telecommunications System (UMTS), Wideband Code Division Multiple Access (W-CDM), High Speed Packet Access (HSPA), HSPA Plus (HSPA+), Time Division-Code Division Multiple Access (TD-CDMA), Time Division-Synchronous Code Division Multiple Access (TD-SCDMA), LTE LAA, MuLTEfire, UMTS Terrestrial Radio Access (UTRA), Evolved UTRA (E-UTRA), Evolution-Data Optimized or Evolution-Data Only (EV-DO), Advanced Mobile Phone System (AMPS), Digital AMPS (D-AMPS), Total Access Communication System/Extended Total Access Communication System (TACS/ETACS), Push-to-talk (PTT), Mobile Telephone System (MTS), Improved Mobile Telephone System (IMTS), Advanced Mobile Telephone System (AMTS), Cellular Digital Packet Data (CDPD), DataTAC, Integrated Digital Enhanced Network (iDEN), Personal Digital Cellular (PDC), Personal Handy-phone System (PHS), Wideband Integrated Digital Enhanced Network (WiDEN), iBurst, Unlicensed Mobile Access (UMA), also referred to as also referred to as 3GPP Generic Access Network, or GAN standard), Bluetooth®, Bluetooth Low Energy (BLE), IEEE 802.15.4 based protocols (e.g., IPv6 over Low power Wireless Personal Area Networks (6LoWPAN), WirelessHART, MiWi, Thread, 802.11a, etc.) WiFi-direct, ANT/ANT+, ZigBee, Z-Wave, 3GPP device-to-device (D2D) or Proximity Services (ProSe), Universal Plug and Play (UPnP), Low-Power Wide-Area-Network (LPWAN), Long Range Wide Area Network (LoRA) or LoRaWAN™ developed by Semtech and the LoRa Alliance, Sigfox, Wireless Gigabit Alliance (WiGig) standard, Worldwide Interoperability for Microwave Access (WiMAX), mmWave standards in general (e.g., wireless systems operating at 10-300 GHz and above such as WiGig, IEEE 802.11ad, IEEE 802.11ay, etc.), V2X communication technologies (including 3GPP C-V2X), Dedicated Short Range Communications (DSRC) communication systems such as Intelligent-Transport-Systems (ITS) including the European ITS-G5, ITS-G5B, ITS-G5C, etc. In addition to the standards listed above, any number of satellite uplink technologies may be used for purposes of the present disclosure including, for example, radios compliant with standards issued by the International Telecommunication Union (ITU), or the European Telecommunications Standards Institute (ETSI), among others. The examples provided herein are thus understood as being applicable to various other communication technologies, both existing and not yet formulated.

[0198] The term "access network" refers to any network, using any combination of radio technologies, RATs, and/or communication protocols, used to connect user devices and service providers. In the context of WLANs, an "access network" is an IEEE 802 local area network (LAN) or metropolitan area network (MAN) between terminals and access routers connecting to provider services. The term "access router" refers to router that terminates a medium access control (MAC) service from terminals and forwards user traffic to information servers according to Internet Protocol (IP) addresses.

[0199] The term "SMTC" refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration. The term "SSB" refers to a synchronization signal/Physical Broadcast Channel (SS/PBCH) block, which includes a Primary Syncrhonization Signal (PSS), a Secondary Syncrhonization Signal (SSS), and a PBCH. The term "a "Primary Cell" refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure. The term "Primary SCG Cell" refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation. The term "Secondary Cell" refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA. The term "Secondary Cell Group" refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC. The term "Serving Cell" refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell. The term "serving cell" or "serving cells" refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA. The term "Special Cell" refers to the PCell of the MCG or the PSCell of the SCG for DC operation; otherwise, the term "Special Cell" refers to the Pcell.

[0200] The term "A1 policy" refers to a type of declarative policies expressed using formal statements that enable the non-RT RIC function in the SMO to guide the near-RT RIC function, and hence the RAN, towards better fulfilment of the RAN intent.

[0201] The term "A1 Enrichment information" refers to information utilized by near-RT RIC that is collected or derived at SMO/non-RT RIC either from non-network data sources or from network functions themselves.

[0202] The term "A1-Policy Based Traffic Steering Process Mode" refers to an operational mode in which the Near-RT RIC is configured through A1 Policy to use Traffic Steering Actions to ensure a more specific notion of network performance (for example, applying to smaller groups of E2 Nodes and UEs in the RAN) than that which it ensures in the Background Traffic Steering.

[0203] The term "Background Traffic Steering Processing Mode" refers to an operational mode in which the Near-RT RIC is configured through O1 to use Traffic Steering Actions to ensure a general background network performance which applies broadly across E2 Nodes and UEs in the RAN.

[0204] The term "Baseline RAN Behavior" refers to the default RAN behavior as configured at the E2 Nodes by SMO

[0205] The term "E2" refers to an interface connecting the Near-RT RIC and one or more O-CU-CPs, one or more O-CU-UPs, one or more O-DUs, and one or more O-eNBs.

[0206] The term "E2 Node" refers to a logical node terminating E2 interface. In this version of the specification, ORAN nodes terminating E2 interface are: for NR access: O-CU-CP, O-CU-UP, O-DU or any combination; and for E-UTRA access: O-eNB.

[0207] The term "Intents", in the context of O-RAN systems/implementations, refers to declarative policy to steer or guide the behavior of RAN functions, allowing the RAN function to calculate the optimal result to achieve stated objective.

[0208] The term "O-RAN non-real-time RAN Intelligent Controller" or "non-RT RIC" refers to a logical function that enables non-real-time control and optimization of RAN elements and resources, AI/ML workflow including model training and updates, and policy-based guidance of applications/features in Near-RT RIC.

[0209] The term "Near-RT RIC" or "O-RAN near-real-time RAN Intelligent Controller" refers to a logical function that enables near-real-time control and optimization of RAN elements and resources via fine-grained (e.g., UE basis, Cell basis) data collection and actions over E2 interface.

[0210] The term "O-RAN Central Unit" or "O-CU" refers to a logical node hosting RRC, SDAP and PDCP protocols.

[0211] The term "O-RAN Central Unit-Control Plane" or "O-CU-CP" refers to a logical node hosting the RRC and the control plane part of the PDCP protocol.

[0212] The term "O-RAN Central Unit-User Plane" or "O-CU-UP" refers to a logical node hosting the user plane part of the PDCP protocol and the SDAP protocol

[0213] The term "O-RAN Distributed Unit" or "O-DU" refers to a logical node hosting RLC/MAC/High-PHY layers based on a lower layer functional split.

[0214] The term "O-RAN eNB" or "O-eNB" refers to an eNB or ng-eNB that supports E2 interface.

[0215] The term "O-RAN Radio Unit" or "O-RU" refers to a logical node hosting Low-PHY layer and RF processing based on a lower layer functional split. This is similar to 3GPP's "TRP" or "RRH" but more specific in including the Low-PHY layer (FFT/iFFT, PRACH extraction).

[0216] The term "O1" refers to an interface between orchestration & management entities (Orchestration/NMS) and O-RAN managed elements, for operation and management, by which FCAPS management, Software management, File management and other similar functions shall be achieved.

[0217] The term "RAN UE Group" refers to an aggregations of UEs whose grouping is set in the E2 nodes through E2 procedures also based on the scope of A1 policies. These groups can then be the target of E2 CONTROL or POLICY messages.

[0218] The term "Traffic Steering Action" refers to the use of a mechanism to alter RAN behavior. Such actions include E2 procedures such as CONTROL and POLICY.

[0219] The term "Traffic Steering Inner Loop" refers to the part of the Traffic Steering processing, triggered by the arrival of periodic TS related KPM (Key Performance Measurement) from E2 Node, which includes UE grouping, setting additional data collection from the RAN, as well as selection and execution of one or more optimization actions to enforce Traffic Steering policies.

[0220] The term "Traffic Steering Outer Loop" refers to the part of the Traffic Steering processing, triggered by the near-RT RIC setting up or updating Traffic Steering aware resource optimization procedure based on information from A1 Policy setup or update, A1 Enrichment Information (EI) and/or outcome of Near-RT RIC evaluation, which includes the initial configuration (preconditions) and injection of related A1 policies, Triggering conditions for TS changes.

[0221] The term "Traffic Steering Processing Mode" refers to an operational mode in which either the RAN or the Near-RT RIC is configured to ensure a particular network performance. This performance includes such aspects as cell load and throughput, and can apply differently to different E2 nodes and UEs. Throughout this process, Traffic Steering Actions are used to fulfill the requirements of this configuration.

[0222] The term "Traffic Steering Target" refers to the intended performance result that is desired from the network, which is configured to Near-RT RIC over 01.

[0223] Furthermore, any of the disclosed embodiments and example implementations can be embodied in the form of various types of hardware, software, firmware, middleware, or combinations thereof, including in the form of control logic, and using such hardware or software in a modular or integrated manner. Additionally, any of the software components or functions described herein can be implemented as software, program code, script, instructions, etc., operable to be executed by processor circuitry. These components, functions, programs, etc., can be developed using any suitable computer language such as, for example, Python, PyTorch, NumPy, Ruby, Ruby on Rails, Scala, Smalltalk, Java™, C++, C#, "C", Kotlin, Swift, Rust, Go (or "Golang"), EMCAScript, JavaScript, TypeScript, Jscript, ActionScript, Server-Side JavaScript (SSJS), PHP, Pearl, Lua, Torch/Lua with Just-In Time compiler (LuaJIT), Accelerated Mobile Pages Script (AMPscript), VBScript, JavaServer Pages (JSP), Active Server Pages (ASP), Node.js, ASP.NET, JAMscript, Hypertext Markup Language (HTML), extensible HTML (XHTML), Extensible Markup Language (XML), XML User Interface Language (XUL), Scalable Vector Graphics (SVG), RESTful API Modeling Language (RAML), wiki markup or Wikitext, Wireless Markup Language (WML), Java Script Object Notion (JSON), Apache® MessagePack™, Cascading Stylesheets (CSS), extensible stylesheet language (XSL), Mustache template language, Handlebars template language, Guide Template Language (GTL), Apache® Thrift, Abstract Syntax Notation One (ASN.1), Google® Protocol Buffers (protobuf), Bitcoin Script, EVM® bytecode, Solidity™, Vyper (Python derived), Bamboo, Lisp Like Language (LLL), Simplicity provided by Blockstream™, Rholang, Michelson, Counterfactual, Plasma, Plutus, Sophia, Salesforce® Apex®, and/or any other programming language or development tools including proprietary programming languages and/or development tools. The software code can be stored as a computer- or processor-executable instructions or commands on a physical non-transitory computer-readable medium. Examples of suitable media include RAM, ROM, magnetic media such as a hard-drive or a floppy disk, or an optical medium such as a compact disk (CD) or DVD (digital versatile disk), flash memory, and the like, or any combination of such storage or transmission devices.

Abbreviations

[0224] Unless used differently herein, terms, definitions, and abbreviations may be consistent with terms, definitions, and abbreviations defined in 3GPP TR 21.905 v16.0.0 (2019-06). For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein.

Table X1X Abbreviations:

3GPP Third Generation Partnership Project
4G Fourth Generation
5G Fifth Generation
5GC 5G Core network
AC Application Client
ACK Acknowledgement

-continued

ACID Application Client Identification
AF Application Function
AM Acknowledged Mode
AMBR Aggregate Maximum Bit Rate
AMF Access and Mobility Management Function
AN Access Network
ANR Automatic Neighbour Relation
AP Application Protocol, Antenna Port, Access Point
API Application Programming Interface
APN Access Point Name
ARP Allocation and Retention Priority
ARQ Automatic Repeat Request
AS Access Stratum
ASP Application Service Provider
ASN.1 Abstract Syntax Notation One
AUSF Authentication Server Function
AWGN Additive White Gaussian Noise
BAP Backhaul Adaptation Protocol
BCH Broadcast Channel
BER Bit Error Ratio
BFD Beam Failure Detection
BLER Block Error Rate
BPSK Binary Phase Shift Keying
BRAS Broadband Remote Access Server
BSS Business Support System
BS Base Station
BSR Buffer Status Report
BW Bandwidth
BWP Bandwidth Part
C-RNTI Cell Radio Network Temporary Identity
CA Carrier Aggregation, Certification Authority
CAPEX CAPital EXpenditure
CBRA Contention Based Random Access
CC Component Carrier, Country Code, Cryptographic Checksum
CCA Clear Channel Assessment
CCE Control Channel Element
CCCH Common Control Channel
CE Coverage Enhancement
CDM Content Delivery Network
CDMA Code-Division Multiple Access
CFRA Contention Free Random Access
CG Cell Group
CGF Charging Gateway Function
CHF Charging Function
CI Cell Identity
CID Cell-ID (e.g., positioning method)
CIM Common Information Model
CIR Carrier to Interference Ratio
CK Cipher Key
CM Connection Management, Conditional Mandatory
CMAS Commercial Mobile Alert Service
CMD Command
CMS Cloud Management System
CO Conditional Optional
CoMP Coordinated Multi-Point
CORESET Control Resource Set
COTS Commercial Off-The-Shelf
CP Control Plane, Cyclic Prefix, Connection Point
CPD Connection Point Descriptor
CPE Customer Premise Equipment
CPICH Common Pilot Channel
CQI Channel Quality Indicator
CPU CSI processing unit, Central Processing Unit
C/R Command/Response field bit
CRAN Cloud Radio Access Network, Cloud RAN
CRB Common Resource Block
CRC Cyclic Redundancy Check
CRI Channel-State Information Resource Indicator,
CSI-RS Resource Indicator
C-RNTI Cell RNTI
CS Circuit Switched
CSAR Cloud Service Archive
CSI Channel-State Information
CSI-IM CSI Interference Measurement
CSI-RS CSI Reference Signal
CSI-RSRP CSI reference signal received power

-continued

CSI-RSRQ CSI reference signal received quality
CSI-SINR CSI signal-to-noise and interference ratio
CSMA Carrier Sense Multiple Access
CSMA/CA CSMA with collision avoidance
CSS Common Search Space, Cell-specific Search Space
CTF Charging Trigger Function
CTS Clear-to-Send
CW Codeword
CWS Contention Window Size
D2D Device-to-Device
DC Dual Connectivity, Direct Current
DCI Downlink Control Information
DF Deployment Flavour
DL Downlink
DMTF Distributed Management Task Force
DPDK Data Plane Development Kit
DM-RS, DMRS Demodulation Reference Signal
DN Data network
DNN Data Network Name
DNAI Data Network Access Identifier
DRB Data Radio Bearer
DRS Discovery Reference Signal
DRX Discontinuous Reception
DSL Domain Specific Language. Digital Subscriber Line
DSLAM DSL Access Multiplexer
DwPTS Downlink Pilot Time Slot
E-LAN Ethernet Local Area Network
E2E End-to-End
ECCA extended clear channel assessment, extended CCA
ECCE Enhanced Control Channel Element, Enhanced CCE
ED Energy Detection
EDGE Enhanced Datarates for GSM Evolution (GSM Evolution)
EAS Edge Application Server
EASID Edge Application Server Identification
ECS Edge Configuration Server
ECSP Edge Computing Service Provider
EDN Edge Data Network
EEC Edge Enabler Client
EECID Edge Enabler Client Identification
EES Edge Enabler Server
EESID Edge Enabler Server Identification
EHE Edge Hosting Environment
EGMF Exposure Governance Management Function
EGPRS Enhanced GPRS
EIR Equipment Identity Register
eLAA enhanced Licensed Assisted Access, enhanced LAA
EM Element Manager
eMBB Enhanced Mobile Broadband
EMS Element Management System
eNB evolved NodeB, E-UTRAN Node B
EN-DC E-UTRA-NR Dual Connectivity
EPC Evolved Packet Core
EPDCCH enhanced PDCCH, enhanced Physical Downlink Control Cannel
EPRE Energy per resource element
EPS Evolved Packet System
EREG enhanced REG, enhanced resource element groups
ETSI European Telecommunications Standards Institute
ETWS Earthquake and Tsunami Warning System
eUICC embedded UICC, embedded Universal Integrated Circuit Card
E-UTRA Evolved UTRA
E-UTRAN Evolved UTRAN
EV2X Enhanced V2X
F1AP F1 Application Protocol
F1-C F1 Control plane interface
F1-U F1 User plane interface
FACCH Fast Associated Control CHannel
FACCH/F Fast Associated Control Channel/Full rate
FACCH/H Fast Associated Control Channel/Half rate
FACH Forward Access Channel
FAUSCH Fast Uplink Signalling Channel
FB Functional Block
FBI Feedback Information
FCC Federal Communications Commission
FCCH Frequency Correction CHannel
FDD Frequency Division Duplex
FDM Frequency Division Multiplex

-continued

FDMA Frequency Division Multiple Access
FE Front End
FEC Forward Error Correction
FFS For Further Study
FFT Fast Fourier Transformation
feLAA further enhanced Licensed Assisted Access, further enhanced LAA
FN Frame Number
FPGA Field-Programmable Gate Array
FR Frequency Range
FQDN Fully Qualified Domain Name
G-RNTI GERAN Radio Network Temporary Identity
GERAN GSM EDGE RAN, GSM EDGE Radio Access Network
GGSN Gateway GPRS Support Node
GLONASS GLObal'naya NAvigatsionnaya Sputnikovaya Sistema
(Engl.: Global Navigation Satellite System)
gNB Next Generation NodeB
gNB-CUgNB-centralized unit, Next Generation NodeB centralized unit
gNB-DUgNB-distributed unit, Next Generation NodeB distributed unit
GNSS Global Navigation Satellite System
GPRS General Packet Radio Service
GPSI Generic Public Subscription Identifier
GSM Global System for Mobile Communications, Groupe Spécial Mobile
GTP GPRS Tunneling Protocol
GTP-U GPRS Tunnelling Protocol for User Plane
GTS Go To Sleep Signal (related to WUS)
GUMMEI Globally Unique MME Identifier
GUTI Globally Unique Temporary UE Identity
HARQ Hybrid ARQ, Hybrid Automatic Repeat Request
HANDO Handover
HFN HyperFrame Number
HHO Hard Handover
HLR Home Location Register
HN Home Network
HO Handover
HPLMN Home Public Land Mobile Network
HSDPA High Speed Downlink Packet Access
HSN Hopping Sequence Number
HSPA High Speed Packet Access
HSS Home Subscriber Server
HSUPA High Speed Uplink Packet Access
HTTP Hyper Text Transfer Protocol
HTTPS Hyper Text Transfer Protocol Secure
(https is http/1.1 over SSL, i.e. port 443)
I-Block Information Block
ICCID Integrated Circuit Card Identification
IAB Integrated Access and Backhaul
ICIC Inter-Cell Interference Coordination
ID Identity, identifier
IDFT Inverse Discrete Fourier Transform
IE Information element
IBE In-Band Emission
IEEE Institute of Electrical and Electronics Engineers
IEI Information Element Identifier
IEIDL Information Element Identifier Data Length
IETF Internet Engineering Task Force
IF Infrastructure
IM Interference Measurement, Intermodulation, IP Multimedia
IMC IMS Credentials
IMEI International Mobile Equipment Identity
IMGI International mobile group identity
IMPI IP Multimedia Private Identity
IMPU IP Multimedia PUblic identity
IMS IP Multimedia Subsystem
IMSI International Mobile Subscriber Identity
IoT Internet of Things
IP Internet Protocol
Ipsec IP Security, Internet Protocol Security
IP-CAN IP-Connectivity Access Network
IP-M IP Multicast
IPv4 Internet Protocol Version 4
IPv6 Internet Protocol Version 6
IR Infrared
IS In Sync
IRP Integration Reference Point
ISDN Integrated Services Digital Network
ISIM IM Services Identity Module

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ISO International Organisation for Standardisation
ISP Internet Service Provider
IWF Interworking-Function
I-WLAN Interworking WLAN
Constraint length of the convolutional code, USIM Individual key
kB Kilobyte (1000 bytes)
kbps kilo-bits per second
Kc Ciphering key
Ki Individual subscriber authentication key
KPI Key Performance Indicator
KQI Key Quality Indicator
KSI Key Set Identifier
ksps kilo-symbols per second
KVM Kernel Virtual Machine
L1 Layer 1 (physical layer)
L1-RSRP Layer 1 reference signal received power
L2 Layer 2 (data link layer)
L3 Layer 3 (network layer)
LAA Licensed Assisted Access
LAN Local Area Network
LADN Local Area Data Network
LBT Listen Before Talk
LCM LifeCycle Management
LCR Low Chip Rate
LCS Location Services
LCID Logical Channel ID
LI Layer Indicator
LLC Logical Link Control, Low Layer Compatibility
LPLMN Local PLMN
LPP LTE Positioning Protocol
LSB Least Significant Bit
LTE Long Term Evolution
LWA LTE-WLAN aggregation
LWIP LTE/WLAN Radio Level Integration with IPsec Tunnel
LTE Long Term Evolution
M2M Machine-to-Machine
MAC Medium Access Control (protocol layering context)
MAC Message authentication code (security/encryption context)
MAC-A MAC used for authentication and key agreement
(TSG T WG3 context)
MAC-I MAC used for data integrity of signalling messages
(TSG T WG3 context)
MANO Management and Orchestration
MBMS Multimedia Broadcast and Multicast Service
MBSFN Multimedia Broadcast multicast service
Single Frequency Network
MCC Mobile Country Code
MCG Master Cell Group
MCOT Maximum Channel Occupancy Time
MCS Modulation and coding scheme
MDAF Management Data Analytics Function
MDAS Management Data Analytics Service
MDT Minimization of Drive Tests
ME Mobile Equipment
MeNB master eNB
MER Message Error Ratio
MGL Measurement Gap Length
MGRP Measurement Gap Repetition Period
MIB Master Information Block, Management Information Base
MIMO Multiple Input Multiple Output
MLC Mobile Location Centre
MM Mobility Management
MME Mobility Management Entity
MN Master Node
MNO Mobile Network Operator
MO Measurement Object, Mobile Originated
MPBCH MTC Physical Broadcast CHannel
MPDCCH MTC Physical Downlink Control CHannel
MPDSCH MTC Physical Downlink Shared CHannel
MPRACH MTC Physical Random Access CHannel
MPUSCH MTC Physical Uplink Shared Channel
MPLS MultiProtocol Label Switching
MS Mobile Station
MSB Most Significant Bit
MSC Mobile Switching Centre

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MSI Minimum System Information,
MCH Scheduling Information
MSID Mobile Station Identifier
MSIN Mobile Station Identification Number
MSISDN Mobile Subscriber ISDN Number
MT Mobile Terminated, Mobile Termination
MTC Machine-Type Communications
mMTC massive MTC, massive Machine-Type Communications
MU-MIMO Multi User MIMO
MWUS MTC wake-up signal, MTC WUS
NACK Negative Acknowledgement
NAI Network Access Identifier
NAS Non-Access Stratum, Non-Access Stratum layer
NCT Network Connectivity Topology
NC-JT Non-Coherent Joint Transmission
NEC Network Capability Exposure
NE-DC NR-E-UTRA Dual Connectivity
NEF Network Exposure Function
NF Network Function
NFP Network Forwarding Path
NFPD Network Forwarding Path Descriptor
NFV Network Functions Virtualization
NFVI NFV Infrastructure
NFVO NFV Orchestrator
NG Next Generation, Next Gen
NGEN-DC NG-RAN E-UTRA-NR Dual Connectivity
NM Network Manager
NMS Network Management System
N-PoP Network Point of Presence
NMIB, N-MIB Narrowband MIB
NPBCH Narrowband Physical Broadcast CHannel
NPDCCH Narrowband Physical Downlink Control CHannel
NPDSCH Narrowband Physical Downlink Shared CHannel
NPRACH Narrowband Physical Random Access CHannel
NPUSCH Narrowband Physical Uplink Shared CHannel
NPSS Narrowband Primary Synchronization Signal
NSSS Narrowband Secondary Synchronization Signal
NR New Radio, Neighbour Relation
NRF NF Repository Function
NRS Narrowband Reference Signal
NS Network Service
NSA Non-Standalone operation mode
NSD Network Service Descriptor
NSR Network Service Record
NSSAI Network Slice Selection Assistance Information
S-NNSAI Single-NSSAI
NSSF Network Slice Selection Function
NW Network
NWUS Narrowband wake-up signal, Narrowband WUS
NZP Non-Zero Power
O&M Operation and Maintenance
ODU2 Optical channel Data Unit - type 2
OFDM Orthogonal Frequency Division Multiplexing
OFDMA Orthogonal Frequency Division Multiple Access
OOB Out-of-Band
OOS Out of Sync
OPEX OPerating EXpense
OSI Other System Information
OSS Operations Support System
OTA over-the-air
PAPR Peak-to-Average Power Ratio
PAR Peak to Average Ratio
PBCH Physical Broadcast Channel
PC Power Control, Personal Computer
PCC Primary Component Carrier, Primary CC
PCell Primary Cell
PCI Physical Cell ID, Physical Cell Identity
PCEF Policy and Charging Enforcement Function
PCF Policy Control Function
PCRFPolicy Control and Charging Rules Function
PDCP Packet Data Convergence Protocol,
Packet Data Convergence Protocol layer
PDCCH Physical Downlink Control Channel
PDCP Packet Data Convergence Protocol
PDN Packet Data Network, Public Data Network
PDSCH Physical Downlink Shared Channel

-continued

PDU Protocol Data Unit
PEI Permanent Equipment Identifiers
PFD Packet Flow Description
P-GW PDN Gateway
PHICH Physical hybrid-ARQ indicator channel
PHY Physical layer
PLMN Public Land Mobile Network
PIN Personal Identification Number
PM Performance Measurement
PMI Precoding Matrix Indicator
PNF Physical Network Function
PNFD Physical Network Function Descriptor
PNFR Physical Network Function Record
POC PTT over Cellular
PP, PTP Point-to-Point
PPP Point-to-Point Protocol
PRACH Physical RACH
PRB Physical resource block
PRG Physical resource block group
ProSe Proximity Services, Proximity-Based Service
PRS Positioning Reference Signal
PRR Packet Reception Radio
PS Packet Services
PSBCH Physical Sidelink Broadcast Channel
PSDCH Physical Sidelink Downlink Channel
PSCCH Physical Sidelink Control Channel
PSSCH Physical Sidelink Shared Channel
PSCell Primary SCell
PSS Primary Synchronization Signal
PSTN Public Switched Telephone Network
PT-RS Phase-tracking reference signal
PTT Push-to-Talk
PUCCH Physical Uplink Control Channel
PUSCH Physical Uplink Shared Channel
QAM Quadrature Amplitude Modulation
QCI QoS class of identifier
QCL Quasi co-location
QFI QoS Flow ID, QoS Flow Identifier
QoS Quality of Service
QPSK Quadrature (Quarternary) Phase Shift Keying
QZSS Quasi-Zenith Satellite System
RA-RNTI Random Access RNTI
RAB Radio Access Bearer, Random Access Burst
RACH Random Access Channel
RADIUS Remote Authentication Dial In User Service
RAN Radio Access Network
RAND RANDom number (used for authentication)
RAR Random Access Response
RAT Radio Access Technology
RAU Routing Area Update
RB Resource block, Radio Bearer
RBG Resource block group
REG Resource Element Group
Rel Release
REQ REQuest
RF Radio Frequency
RI Rank Indicator
RIV Resource indicator value
RL Radio Link
RLC Radio Link Control, Radio Link Control layer
RLC AM RLC Acknowledged Mode
RLC UM RLC Unacknowledged Mode
RLF Radio Link Failure
RLM Radio Link Monitoring
RLM-RS Reference Signal for RLM
RM Registration Management
RMC Reference Measurement Channel
RMSI Remaining MSI, Remaining Minimum System Information
RN Relay Node
RNC Radio Network Controller
RNL Radio Network Layer
RNTI Radio Network Temporary Identifier
ROHC RObust Header Compression
RRC Radio Resource Control, Radio Resource Control layer
RRM Radio Resource Management
RS Reference Signal

-continued

RSRP Reference Signal Received Power
RSRQ Reference Signal Received Quality
RSSI Received Signal Strength Indicator
RSU Road Side Unit
RSTD Reference Signal Time difference
RTP Real Time Protocol
RTS Ready-To-Send
RTT Round Trip Time
Rx Reception, Receiving, Receiver
S1AP S1 Application Protocol
S1-MMES1 for the control plane
S1-U S1 for the user plane
S-GW Serving Gateway
S-RNTI SRNC Radio Network Temporary Identity
S-TMSI SAE Temporary Mobile Station Identifier
SA Standalone operation mode
SAE System Architecture Evolution
SAP Service Access Point
SAPD Service Access Point Descriptor
SAPI Service Access Point Identifier
SCC Secondary Component Carrier, Secondary CC
SCell Secondary Cell
SCEF Service Capability Exposure Function
SC-FDMA Single Carrier Frequency Division Multiple Access
SCG Secondary Cell Group
SCM Security Context Management
SCS Subcarrier Spacing
SCTP Stream Control Transmission Protocol
SDAP Service Data Adaptation Protocol,
Service Data Adaptation Protocol layer
SDL Supplementary Downlink
SDNF Structured Data Storage Network Function
SDP Session Description Protocol
SDSF Structured Data Storage Function
SDU Service Data Unit
SEAF Security Anchor Function
SeNB secondary eNB
SEPP Security Edge Protection Proxy
SFI Slot format indication
SFTD Space-Frequency Time Diversity, SFN and frame timing difference
SFN System Frame Number
SgNB secondary gNB
SGSN Serving GPRS Support Node
S-GW Serving Gateway
SI System Information
SI-RNTI System Information RNTI
SIB System Information Block
SIM Subscriber Identity Module
SIP Session Initiated Protocol
SiP System in Package
SL Sidelink
SLA Service Level Agreement
SM Session Management
SMF Session Management Function
SMS Short Message Service
SMSF SMS Function
SMTC SSB-based Measurement Timing Configuration
SN Secondary Node, Sequence Number
SoC System on Chip
SON Self-Organizing Network
SpCell Special Cell
SP-CSI-RNTI Semi-Persistent CSI RNTI
SPS Semi-Persistent Scheduling
SQN Sequence number
SR Scheduling Request
SRB Signalling Radio Bearer
SRS Sounding Reference Signal
SS Synchronization Signal
SSB Synchronization Signal Block
SSID Service Set Identifier
SS/PBCH Block
SSBRI SS/PBCH Block Resource Indicator,
Synchronization Signal Block Resource Indicator
SSC Session and Service Continuity
SS-RSRP Synchronization Signal based Reference Signal
Received Power

-continued

SS-RSRQ	Synchronization Signal based Reference Signal Received Quality
SS-SINR	Synchronization Signal based Signal to Noise and Interference Ratio
SSS	Secondary Synchronization Signal
SSSG	Search Space Set Group
SSSIF	Search Space Set Indicator
SST	Slice/Service Types
SU-MIMO	Single User MIMO
SUL	Supplementary Uplink
TA	Timing Advance, Tracking Area
TAC	Tracking Area Code
TAG	Timing Advance Group
TAI	Tracking Area Identity
TAU	Tracking Area Update
TB	Transport Block
TBS	Transport Block Size
TBD	To Be Defined
TCI	Transmission Configuration Indicator
TCP	Transmission Communication Protocol
TDD	Time Division Duplex
TDM	Time Division Multiplexing
TDMA	Time Division Multiple Access
TE	Terminal Equipment
TEID	Tunnel End Point Identifier
TFT	Traffic Flow Template
TMSI	Temporary Mobile Subscriber Identity
TNL	Transport Network Layer
TPC	Transmit Power Control
TPMI	Transmitted Precoding Matrix Indicator
TR	Technical Report
TRP, TRxP	Transmission Reception Point
TRS	Tracking Reference Signal
TRx	Transceiver
TS	Technical Specifications, Technical Standard
TTI	Transmission Time Interval
Tx	Transmission, Transmitting, Transmitter
U-RNTI	UTRAN Radio Network Temporary Identity
UART	Universal Asynchronous Receiver and Transmitter
UCI	Uplink Control Information
UE	User Equipment
UDM	Unified Data Management
UDP	User Datagram Protocol
USDF	Unstructured Data Storage Network Function
UICC	Universal Integrated Circuit Card
UL	Uplink
UM	Unacknowledged Mode
UML	Unified Modelling Language
UMTS	Universal Mobile Telecommunications System
UP	User Plane
UPF	User Plane Function
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
URLLC	Ultra-Reliable and Low Latency
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
USS	UE-Specific search space
UTRA	UMTS Terrestrial Radio Access
UTRAN	Universal Terrestrial Radio Access Network
UwPTS	Uplink Pilot Time Slot
V2I	Vehicle-to-Infrastruction
V2P	Vehicle-to-Pedestrian
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-everything
VIM	Virtualized Infrastructure Manager
VL	Virtual Link,
VLAN	Virtual LAN, Virtual Local Area Network
VM	Virtual Machine
VNF	Virtualized Network Function
VNFFG	VNF Forwarding Graph
VNFFGD	VNF Forwarding Graph Descriptor
VNFM	VNF Manager
VoIP	Voice-over-IP, Voice-over-Internet Protocol
VPLMN	Visited Public Land Mobile Network
VPN	Virtual Private Network
VRB	Virtual Resource Block

-continued

WiMAX	Worldwide Interoperability for Microwave Access
WLAN	Wireless Local Area Network
WMAN	Wireless Metropolitan Area Network
WPAN	Wireless Personal Area Network
X2-C	X2-Control plane
X2-U	X2-User plane
XML	eXtensible Markup Language
XRES	EXpected user RESponse
XOR	eXclusive OR
ZC	Zadoff-Chu
ZP	Zero Po

[0225] The foregoing description provides illustration and description of various example embodiments, but is not intended to be exhaustive or to limit the scope of embodiments to the precise forms disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments. Where specific details are set forth in order to describe example embodiments of the disclosure, it should be apparent to one skilled in the art that the disclosure can be practiced without, or with variation of, these specific details. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

What is claimed is:

1. An apparatus of a location management function (LMF) comprising:

processing circuitry configured to:

perform inference for artificial intelligence (AI)/machine learning (ML) based positioning;

receive a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics;

send a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication; and

provide location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF; and

a memory to store the trained ML model.

2. The apparatus of claim **1**, wherein an LMF decision to obtain a trained ML model from the NWDAF is based on implementation requirements or model accuracy monitoring by a service consumer.

3. The apparatus of claim **1**, wherein the processing circuitry is further configured to sending a Nnwdaf_MLModelTraining_Subscribe request for UE AI/ML positioning ML model/analytics.

4. The apparatus of claim **1**, wherein the processing circuitry is further configured to indicate data collection initiation by the NWDAF MTLF from a group mobile location center (GMLC) using a Ngmlc_Location_ProvideLocationMeasurement request.

5. The apparatus of claim **4**, wherein the processing circuitry is further configured to receive a Namf_Location_ProvideAIMLPosMeasurement request from the GMLC via the AMF serving the UE or group of UEs.

6. The apparatus of claim **1**, wherein the processing circuitry is further configured to store measurement data collected in the ADRF and sends the ADRF ID plus DataSetTag to the NWDAF MTLF.

7. The apparatus of claim **1**, wherein the processing circuitry is further configured to collect alternative measurement data from the UE or a radio access network (RAN) if the requested measurement data is unavailable.

8. The apparatus of claim **1**, wherein the processing circuitry is further configured to indicate to the AMF to send a Namf_Location_ProvideAIMLPosMeasurement response to a group mobile location center (GMLC) with the location measurement data or ADRF ID plus DataSetTag.

9. The apparatus of claim **8**, wherein the processing circuitry is further configured to provide location measurement data or ADRF ID plus DataSetTag where the input measurement data is stored to the GMLC for transmission to the NWDAF containing MTLF.

10. The apparatus of claim **1**, wherein the processing circuitry is further configured to receive a Nnwdaf_MLModelTraining_Notify for AI/ML positioning analytics from the NWDAF containing MTLF.

11. A non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors of a location management function (LMF) result in performing operations comprising:

performing inference for artificial intelligence (AI)/machine learning (ML) based positioning;

receiving a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics;

sending a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication; and

providing location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF.

12. The non-transitory computer-readable medium of claim **11**, wherein an LMF decision to obtain a trained ML model from the NWDAF is based on implementation requirements or model accuracy monitoring by a service consumer.

13. The non-transitory computer-readable medium of claim **11**, wherein the operations further comprise sending a Nnwdaf_MLModelTraining_Subscribe request for UE AI/ML positioning ML model/analytics.

14. The non-transitory computer-readable medium of claim **11**, wherein the operations further comprise indicating data collection initiation by the NWDAF MTLF from a group mobile location center (GMLC) using a Ngmlc_Location_ProvideLocationMeasurement request.

15. The non-transitory computer-readable medium of claim **14**, wherein the operations further comprise receiving a Namf_Location_ProvideAIMLPosMeasurement request from the GMLC via the AMF serving the UE or group of UEs.

16. The non-transitory computer-readable medium of claim **11**, wherein the operations further comprise storing measurement data collected in the ADRF and sends the ADRF ID plus DataSetTag to the NWDAF MTLF.

17. The non-transitory computer-readable medium of claim **11**, wherein the operations further comprise collect alternative measurement data from the UE or a radio access network (RAN) if the requested measurement data is unavailable.

18. The non-transitory computer-readable medium of claim **11**, wherein the operations further comprise indicating to the AMF to send a Namf_Location_ProvideAIMLPosMeasurement response to a group mobile location center (GMLC) with the location measurement data or ADRF ID plus DataSetTag.

19. The non-transitory computer-readable medium of claim **18**, wherein the operations further comprise providing location measurement data or ADRF ID plus DataSetTag where the input measurement data is stored to the GMLC for transmission to the NWDAF containing MTLF.

20. A method comprising:

performing, by one or more processors associated with a location management function (LMF), inference for artificial intelligence (AI)/machine learning (ML) based positioning;

receiving a trained ML model from a network data analytics function (NWDAF) containing a model training logic function (MTLF) for user equipment (UE) positioning analytics;

sending a location measurement data request to an access and mobility management function (AMF) with an AI/ML positioning indication; and

providing location measurement data or an analytics data repository function (ADRF) ID plus DataSetTag to the AMF.

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